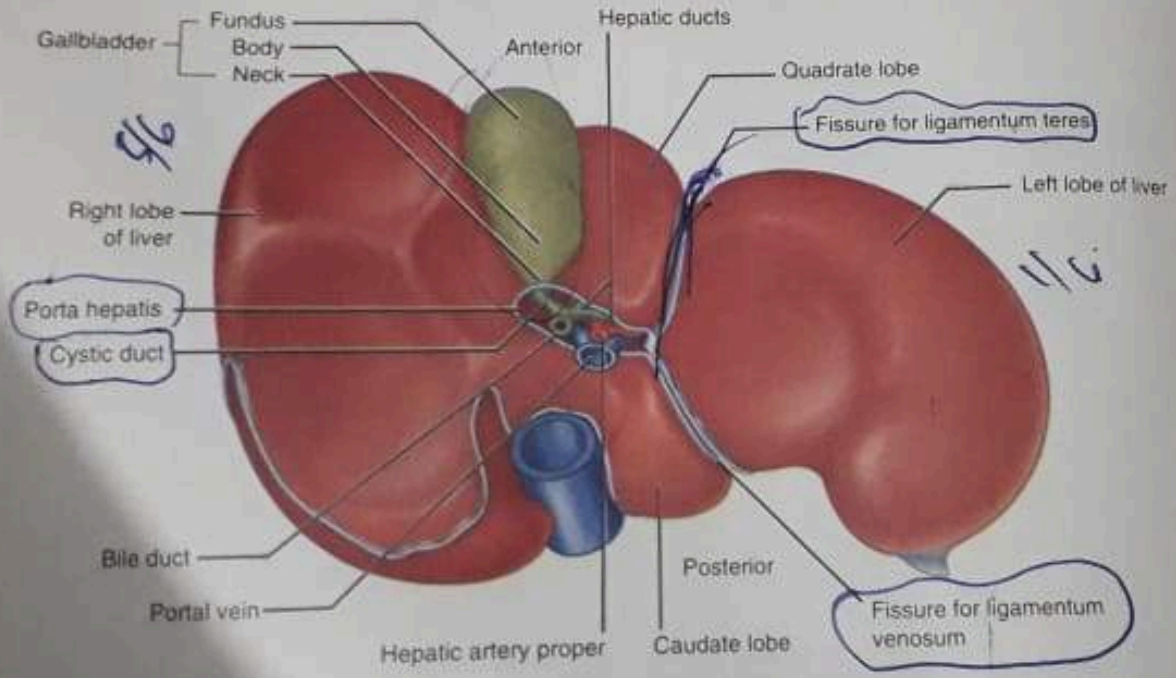
esophagus.
- right anterior part of the stomach.
- superior part of the duodenum.
- lesser omentum.
- gallbladder.

exit point for the hepatic ducts (Fig. 4.95).

Associated ligaments

The liver is attached to the anterior abdominal wall by the **falciform ligament** and, except for a small area of the



A



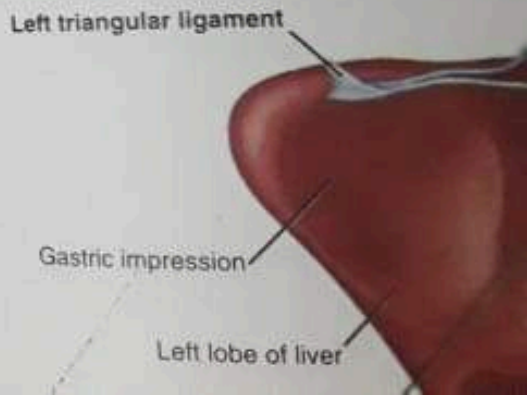
diaphragmatic
toneum between the liver

- The anterior boundary of the bare area is indicated
a reflection of peritoneum—the anterior
ligament.
- The posterior boundary of the bare area is indicated
a reflection of peritoneum—the posterior
ligament.
- Where the coronary ligaments come together
they form the right and left triangular ligaments.

Lobes

The liver is divided into right and left lobes by fossa
gallbladder and the inferior vena cava (Fig. 4.95
right lobe of the liver is the largest lobe, whereas
lobe of the liver is smaller. The quadrate and
lobes are described as arising from the right lobe of
but functionally are distinct.

Falciform ligamen



mucosal fold come together at their end, forming ridges. ...ntiation from the ileum, continues into each flap, ...ming a structure. Possible functions of the ileocecal fold ...de preventing reflux from the cecum to the ileum, and ...ulating the passage of contents from the ileum to the ...um.

The arterial supply to the ileum (Fig. 4.68) includes:

- ...jejunal arteries from the superior mesenteric artery, and
- ...ar ileal branch from the ileocolic artery (from the supe... mesenteric artery).

In the clinic

Epithelial transition between the abdominal esophagus and stomach

The gastroesophageal junction is demarcated by a transition from one epithelial type to another epithelial type. In some people, the histological junction does not lie at the physiological gastroesophageal junction but is in the lower one-third of the esophagus. This may predispose these people to esophageal ulceration and is also associated with an increased risk of adenocarcinoma.

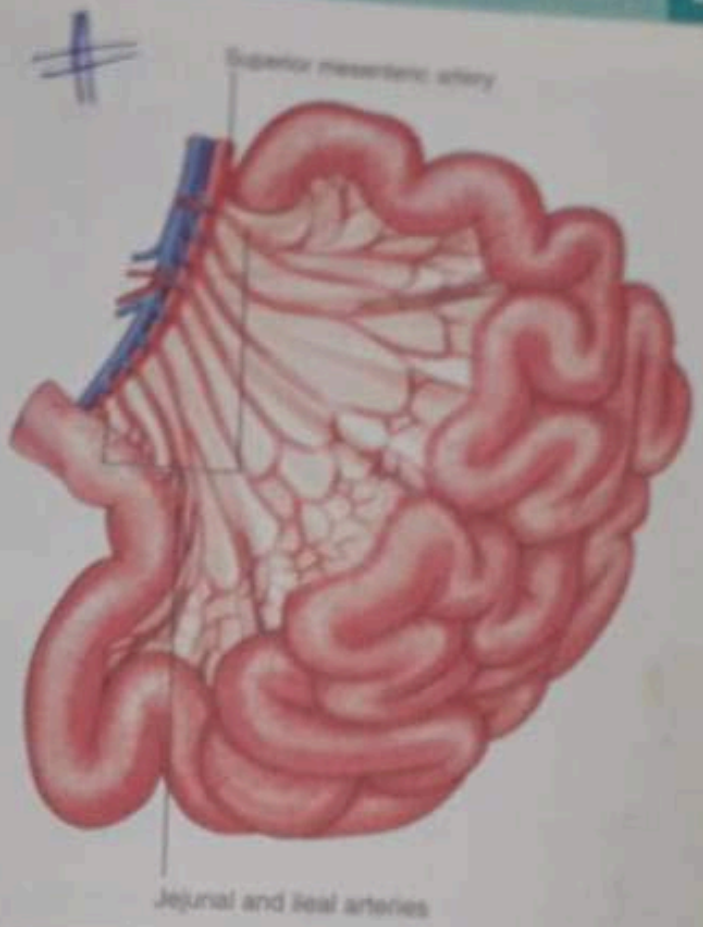


Fig. 4.68 Arterial supply to the ileum.





Fig. 4.83 Right and left colic flexures.

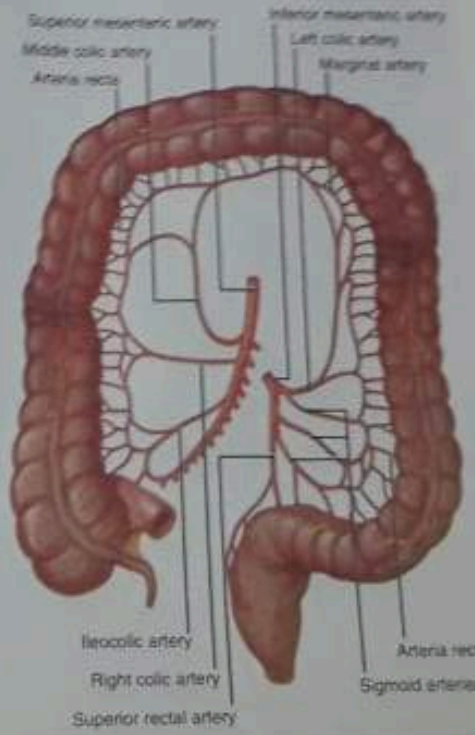
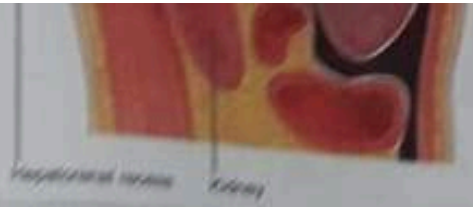
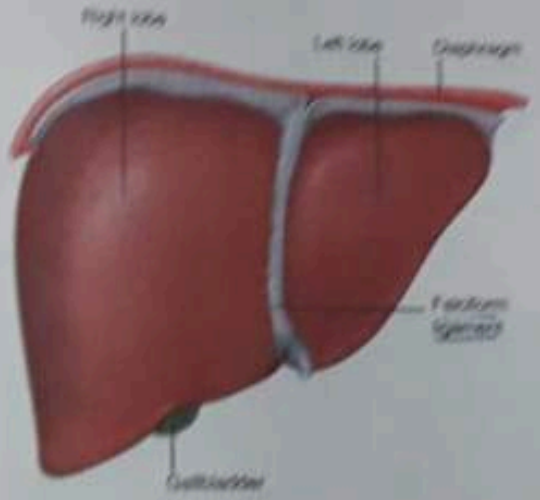


Fig. 4.84 Arterial supply to the colon.



intertubercular plane





PCO_2	46 mm Hg	50 mm Hg ?
pH	7.4	7.0
Proteins	2 g/dl (5 mEq/L)	16 g/dl (40 mEq/L)

Figure 4-1. Chemical compositions of extracellular and intracellular fluids. The question mark indicates that precise values for intracellular fluid are unknown. The red line indicates the cell membrane.

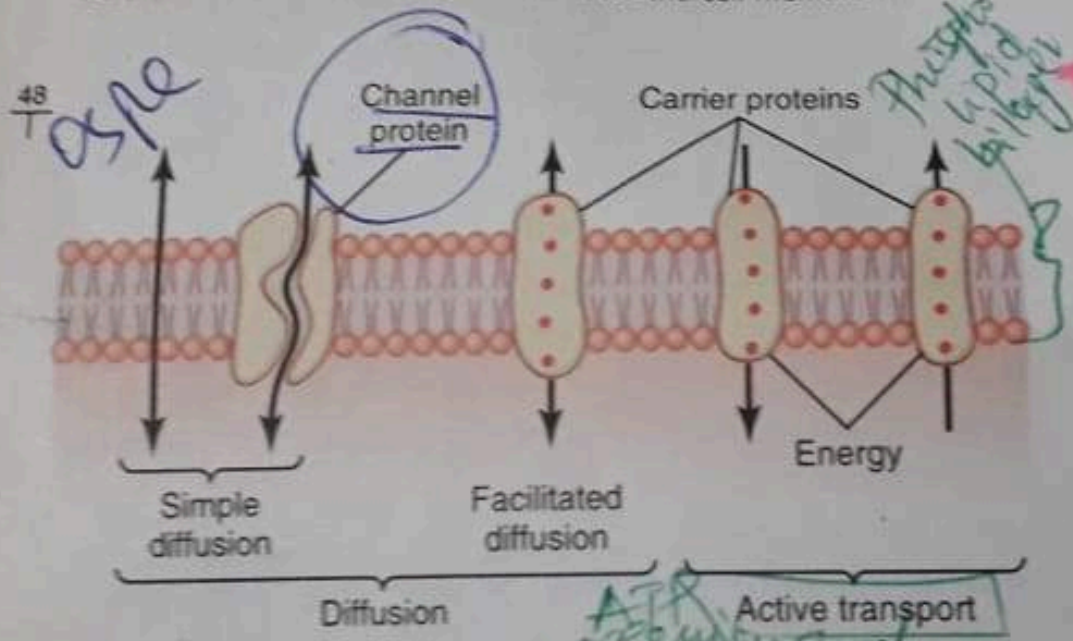


Figure 4-2. Transport pathways through the cell membrane and the basic mechanisms of transport.

direction, then another, then another, and so forth, randomly bouncing thousands of times each second. This

cules or ions can

Facilitated diffusion requires interaction of a protein. The carrier protein aids passage of the molecules or ions through the membrane by binding chemically with them and shuttling them through the membrane in this form.

Simple diffusion can occur through the membrane by two pathways: (1) through the lipid bilayer if the diffusing substance is lipid soluble, (2) through watery channels that penetrate through some of the large transport proteins, as to the left in Figure 4-2.

Diffusion of Lipid-Soluble Substances Through the Lipid Bilayer. An important factor that determines how rapidly a substance diffuses through the lipid bilayer is its *lipid solubility* of the substance. For example, the solubilities of oxygen, nitrogen, carbon dioxide, and alcohols are high, and all these substances can dissolve in the lipid bilayer and diffuse through the membrane in the same manner that diffusion occurs in a watery solution. The rate of diffusion of these substances through the membrane is directly proportional to its lipid solubility. Especially, oxygen can be transported in this way

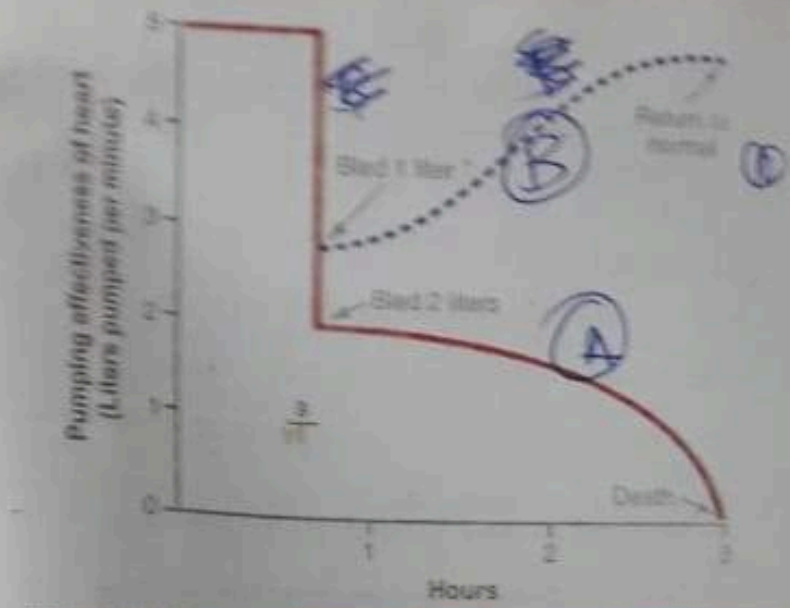
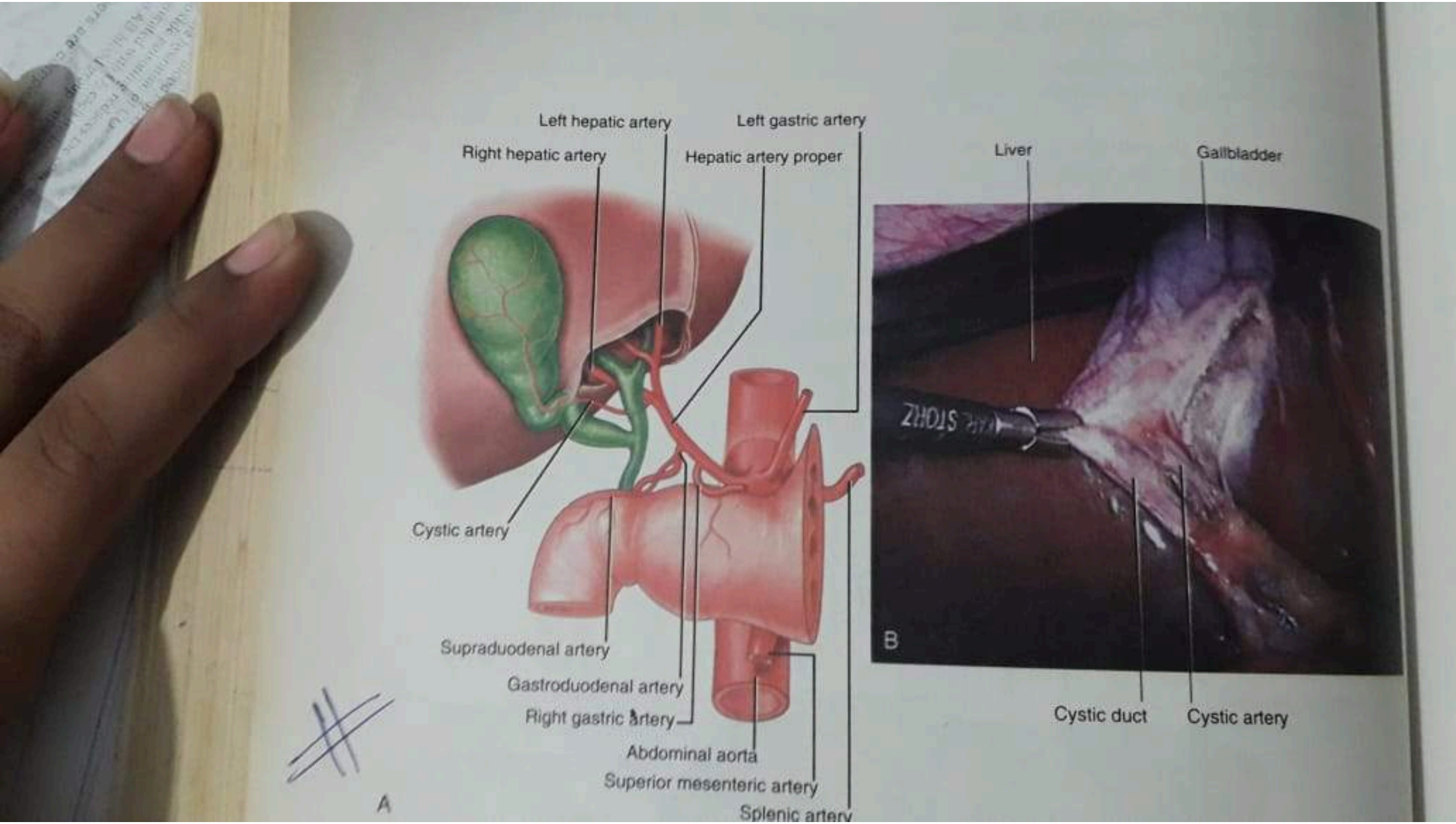


Figure 1-4.

that the temperature control system is much more effective than the baroreceptor pressure control system.

Positive Feedback Can Sometimes C
Vicious Cycles and Death


Left hepatic artery
Left gastric artery
Right hepatic artery
Hepatic artery proper
Cystic artery
Supraduodenal artery
Gastroduodenal artery
Right gastric artery
Abdominal aorta
Superior mesenteric artery
Splenic artery
A
Liver
Gallbladder
B
Cystic duct
Cystic artery

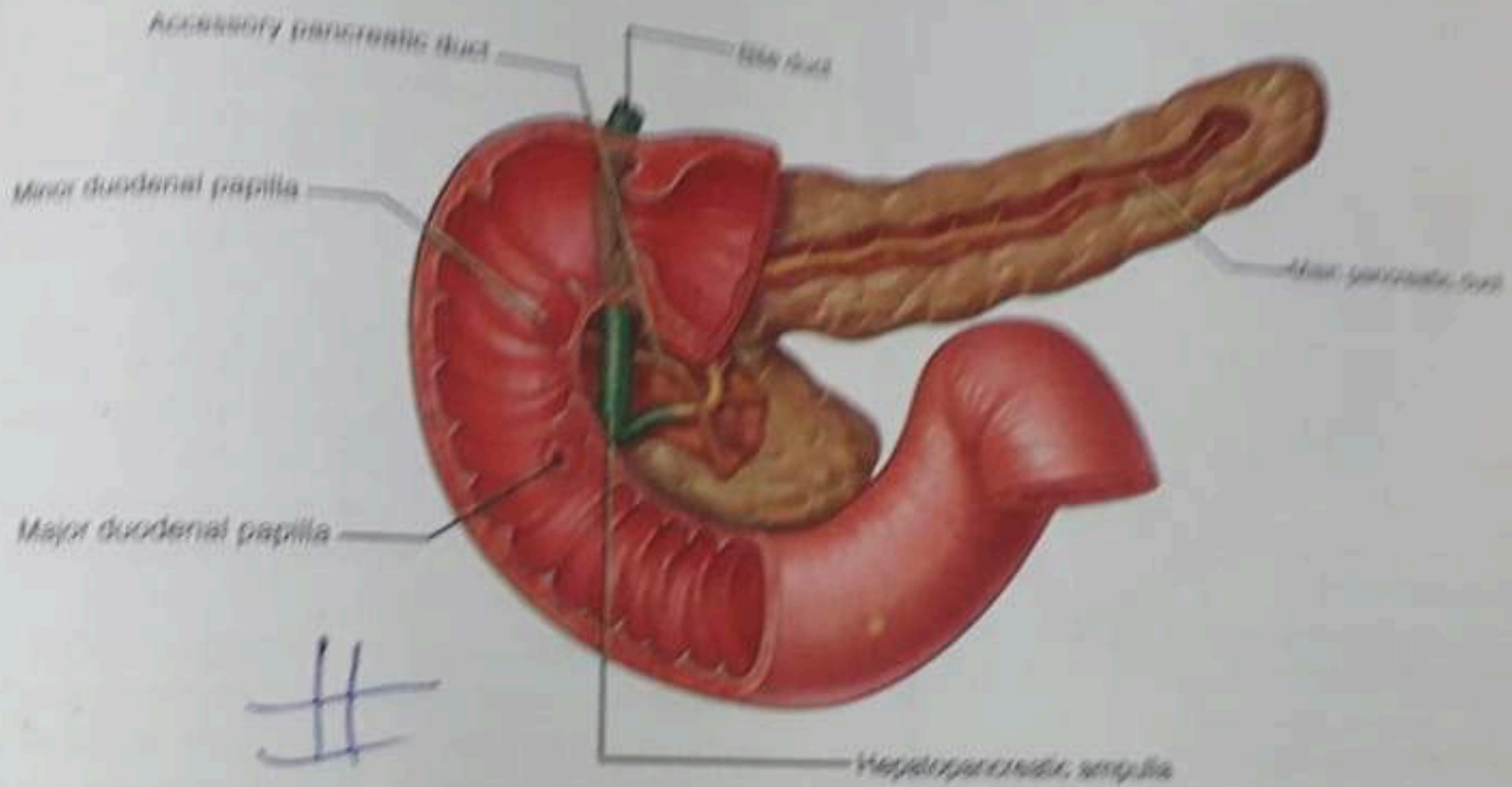


Fig. 4.30 Pancreatic duct system.

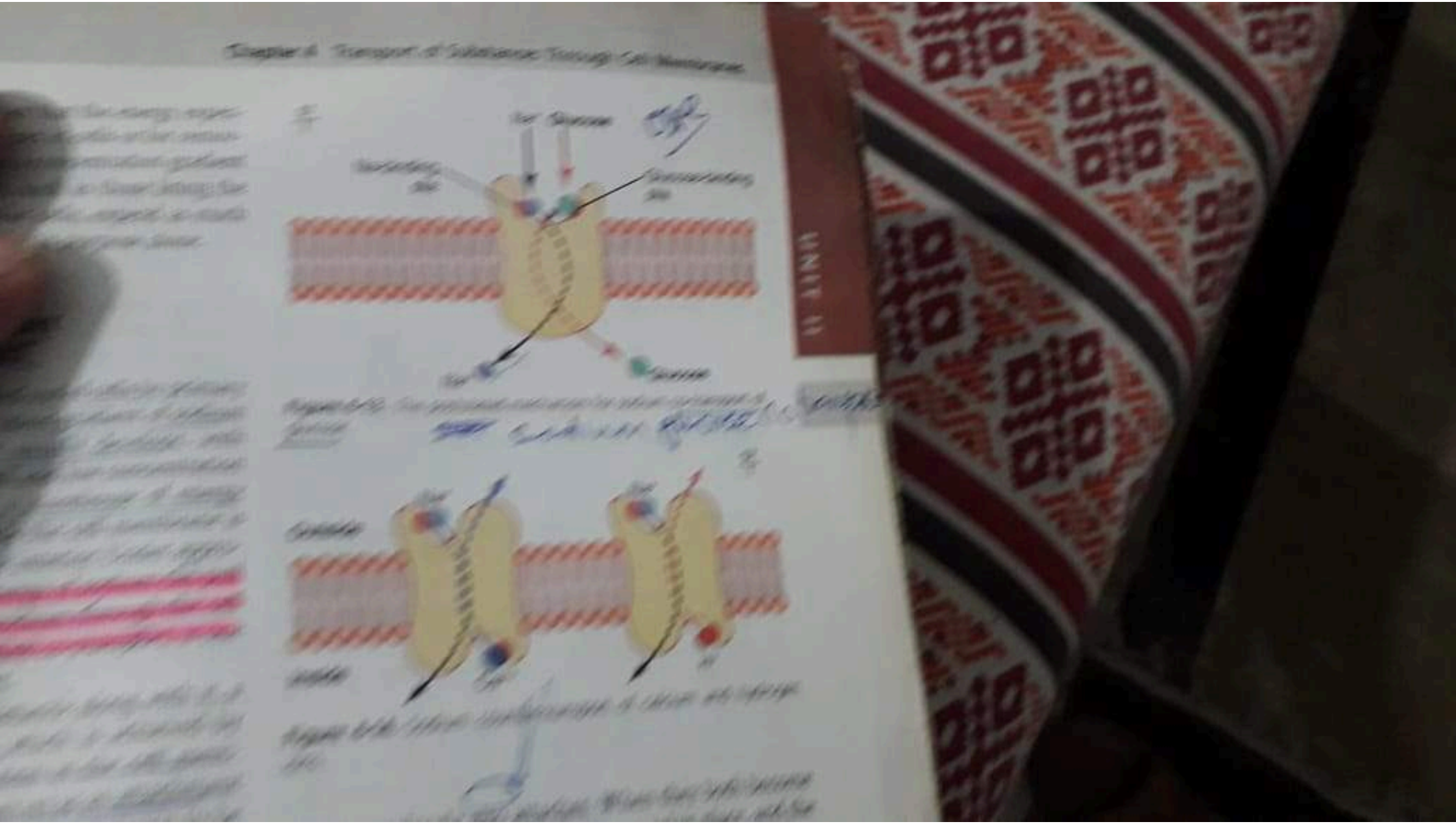




Figure 6-2. An electron micrograph of muscle myofibrils showing the detailed organization of actin and myosin filaments. Note the mitochondria lying between the myofibrils. (From Fawcett DW: The Cell. Philadelphia, WB Saunders, 1981.)

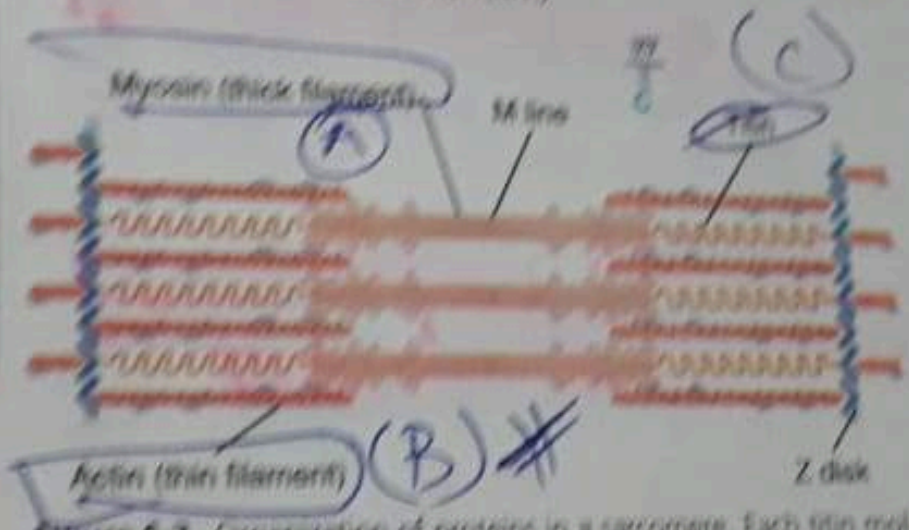


Figure 6-3. Organization of proteins in a sarcomere. Each titin molecule extends from the Z disk to the M line. Part of the titin molecule is closely associated with the myosin thick filament, whereas the rest of the molecule is springy and changes length as the sarcomere contracts and relaxes.

filamentous, it is very *springy*. These springy titin molecules act as a framework that holds the myosin and actin filaments in place so that the contractile machinery of the



Figure 6-4. Sarcoplasmic reticulum in the extracellular spaces between the myofibrils, showing a longitudinal system paralleling the myofibrils. Also shown in cross section are T tubules (arrows) that lead to the exterior of the fiber membrane and are important in conducting the electrical signal into the center of the muscle fiber. (From Fawcett DW: The Cell. Philadelphia, WB Saunders, 1981.)

Sarcoplasmic Reticulum Is a Specialized Endoplasmic Reticulum of Skeletal Muscle. Also in the sarcoplasm surrounding the myofibrils of each muscle fiber is an extensive reticulum (Figure 6-4), called the *sarcoplasmic reticulum*. This reticulum has a special organization that is extremely important in regulating calcium ion release, and reuptake and therefore muscle contraction, as discussed in Chapter 7. The rapidly contracting types of muscle fibers have especially extensive sarcoplasmic reticula.

GENERAL MECHANISM OF MUSCLE CONTRACTION

The initiation and execution of muscle contraction occurs in the following sequential steps.



Figure 9-2. Syncytial, interconnecting nature of cardiac muscle fibers.

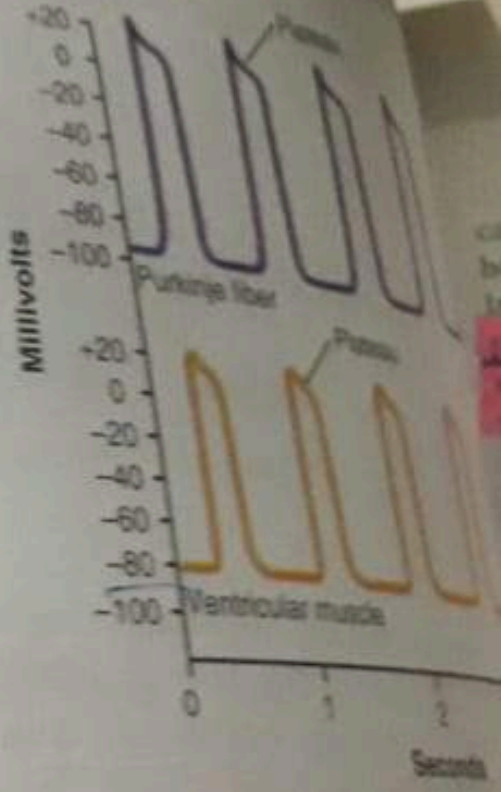


Figure 9-3. Rhythmical action potentials recorded from a Purkinje fiber and from a ventricular muscle fiber, using microelectrodes.

strated easily from one cardiac muscle cell to the next, past the intercalated discs. Thus, cardiac muscle is a syncytium

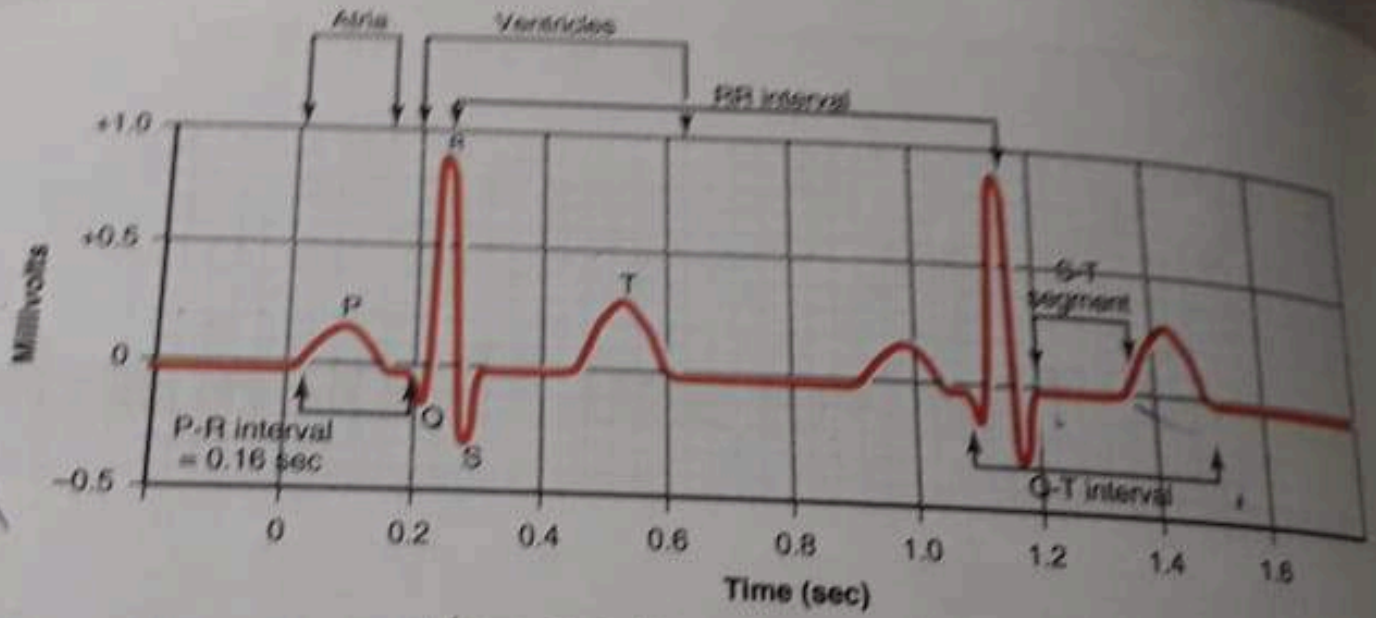


Figure 11-1. Normal electrocardiogram.



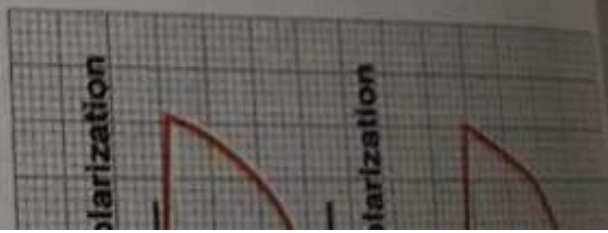




Figure 13-1. Sinus tachycardia (lead II).

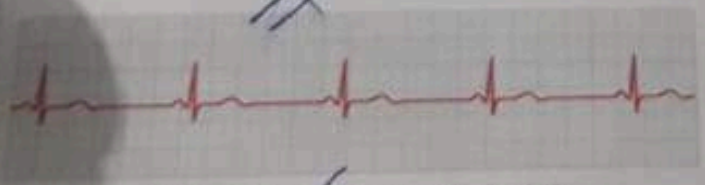


Figure 13-2. Sinus bradycardia (lead III).



Figure 13-3. Sinus arrhythmia as recorded by a cardiotachometer. To the left is the record when the subject was breathing normally; to the right, when the subject was breathing deeply.

SINUS ARRHYTHMIA

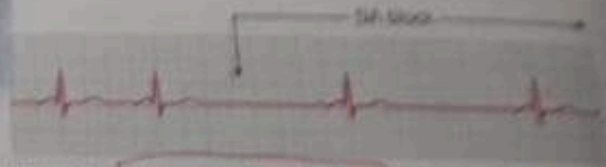


Figure 13-4. Sinoatrial (S-A) nodal block with atrioventricular nodal rhythm during the pause period (lead III).

sudden cessation of P waves, with resultant standstill of the atria. However, the ventricles pick up a new rhythm, with the impulse usually originating spontaneously in the atrioventricular (A-V) node, so the rate of the ventricular QRS-T complex is slowed but not otherwise altered.

ATRIOVENTRICULAR BLOCK

The only means by which impulses ordinarily can pass from the atria into the ventricles is through the A-V *bundle*, also known as the *bundle of His*. Conditions that can either decrease the rate of impulse conduction in this bundle or block the impulse entirely are as follows:

1. *Ischemia of the A-V node or A-V bundle fibers* often delays or blocks conduction from the atria to the ventricles. Coronary insufficiency can cause ad

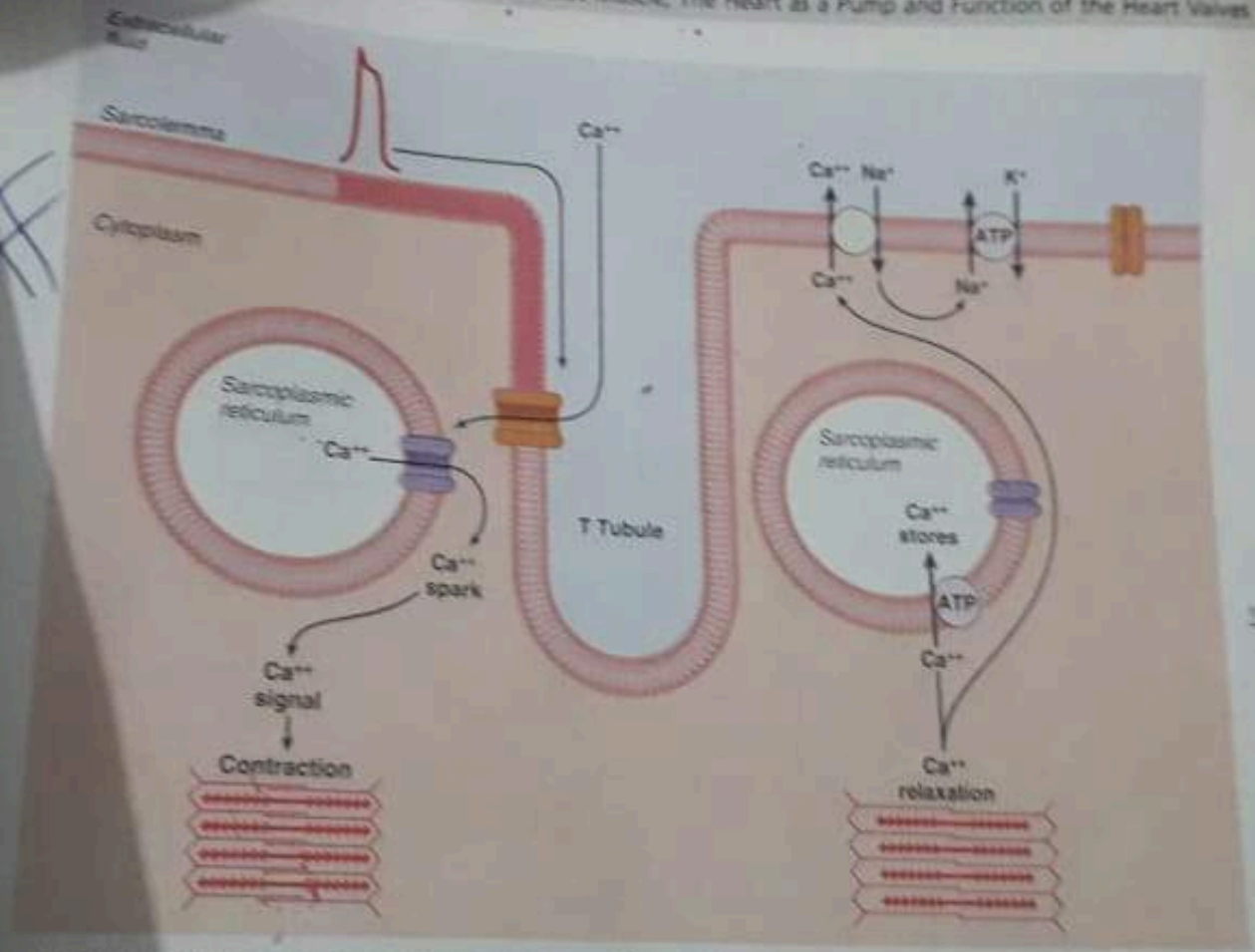


Figure 9-6. Mechanisms of excitation-contraction coupling and relaxation in cardiac muscle. ATP, adenosine triphosphate.

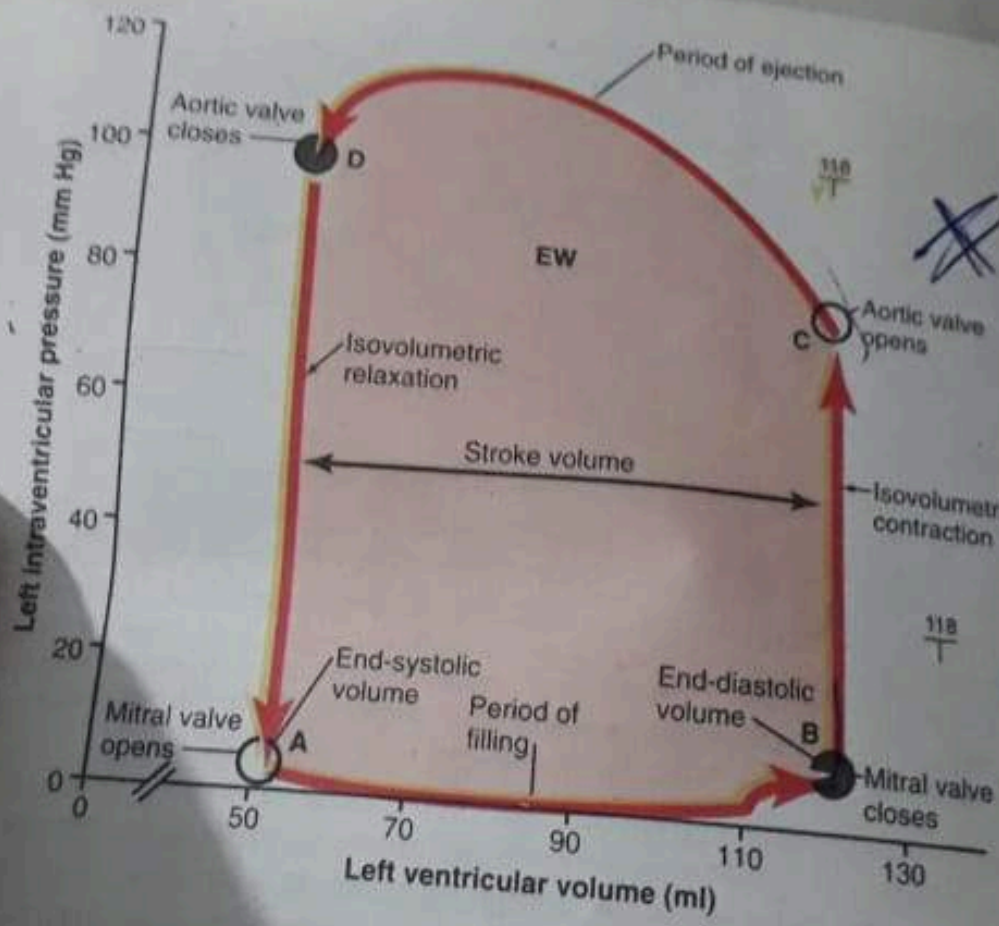


Figure 9-10. This ... onstrating changes in ... pressure during a single ... shaded area represents the ... output by the left ... cycle.

Therefore, in **Figure 9-9** the curve labeled "III," or "period of ejection," traces the changes in volume and systolic pressure during this period of ejection.

Phase IV: Period of ...

For cardiac contraction, the ... ered to be the end-diastolic ...



Figure 13-1. Sinus tachycardia (lead I).



Figure 13-2. Sinus bradycardia (lead II).



Figure 13-3. Sinus arrhythmia as recorded by a cardiotachometer. To the left is the record when the subject was breathing normally; to the right, when the subject was breathing deeply.

SINUS ARRHYTHMIA

Figure 13-3 shows a *cardiotachometer* recording of the

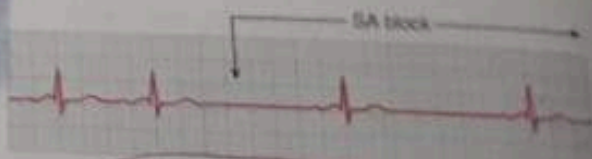


Figure 13-4. Sinoatrial (SA) nodal block, with atrioventricular nodal rhythm during the block period (lead II).

sudden cessation of P waves, with resultant standstill of the atria. However, the ventricles pick up a new rhythm, with the impulse usually originating spontaneously in the atrioventricular (A-V) node, so the rate of the ventricular QRS-T complex is slowed but not otherwise altered.

ATRIOVENTRICULAR BLOCK

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1. *Ischemia of the A-V node or A-V bundle fibers* also delays or blocks conduction from the atria to the ventricles. Coronary insufficiency can cause ab

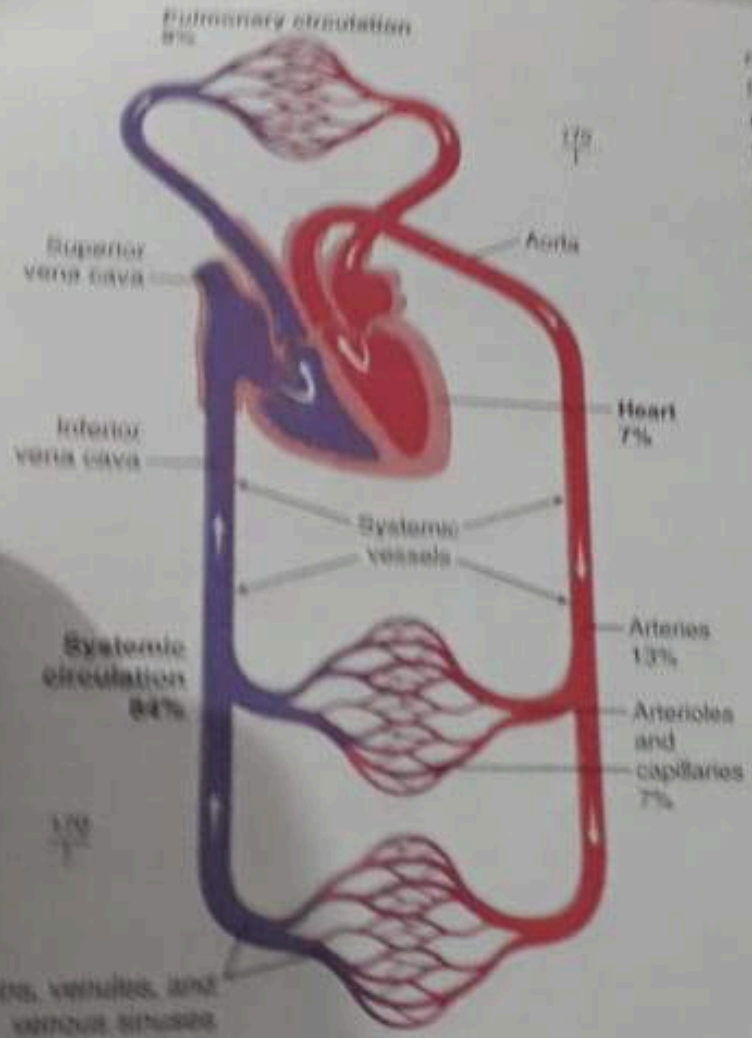


Figure 14-1. Distribution of blood (in percentage of total blood) in the different parts of the circulatory system.

millimeter, the blood remains in the capillaries for only 1/2 to 3 seconds, which is surprising because all substances nutrient food substances and electrolytes that must through the capillary walls must be performed in short time.

Pressures in the Various Portions of the Circulation. Because the heart pumps blood continuously into the aorta, the mean pressure in the aorta is high, normally about 100 mm Hg. Also, because heart pumping is intermittent, the arterial pressure alternates between a high *systolic pressure level* of 120 mm Hg and a *diastolic pressure* level of 80 mm Hg, as shown on the left side of Figure

As the blood flows through the systemic circulation, the mean pressure falls progressively to about 0 mm Hg by the time it reaches the termination of the superior and inferior venae cavae where they empty into the right atrium of the heart.

The pressure in the systemic capillaries varies from as high as 35 mm Hg near the arteriolar ends to as low as 10 mm Hg near the venous ends, but their average "functional" pressure in most vascular beds is about 17 mm Hg, a pressure low enough that little of the plasma filters through the minute *pores* of the capillary walls, although nutrients can *diffuse* easily through these pores to the outlying tissue cells.

Note at the far right side of Figure 14-2 the normal pressures in the different parts of the pulmonary circulation. In the pulmonary arteries, the pressure pulses just as in the aorta, but the pressure is far less pulsatile. The pulmonary artery systolic pressure averages about 25 mm Hg, and the pulmonary diastolic pressure averages about 8 mm Hg, with a mean pressure of only 16 mm Hg.

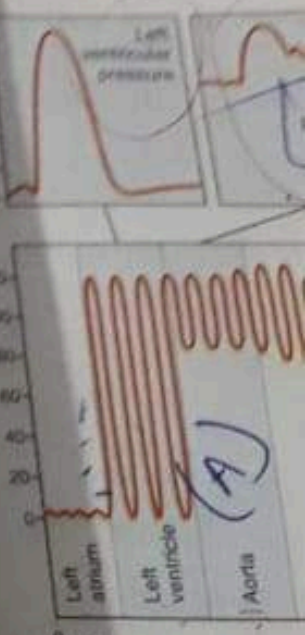


Figure 14-2. Normal blood pressures in the aorta, left ventricle, and left atrium.

tissue continuously
the availability of
the accumulation
waste products, a
directly on the lo
stricting them, t
to that level req
nervous control
nervous system

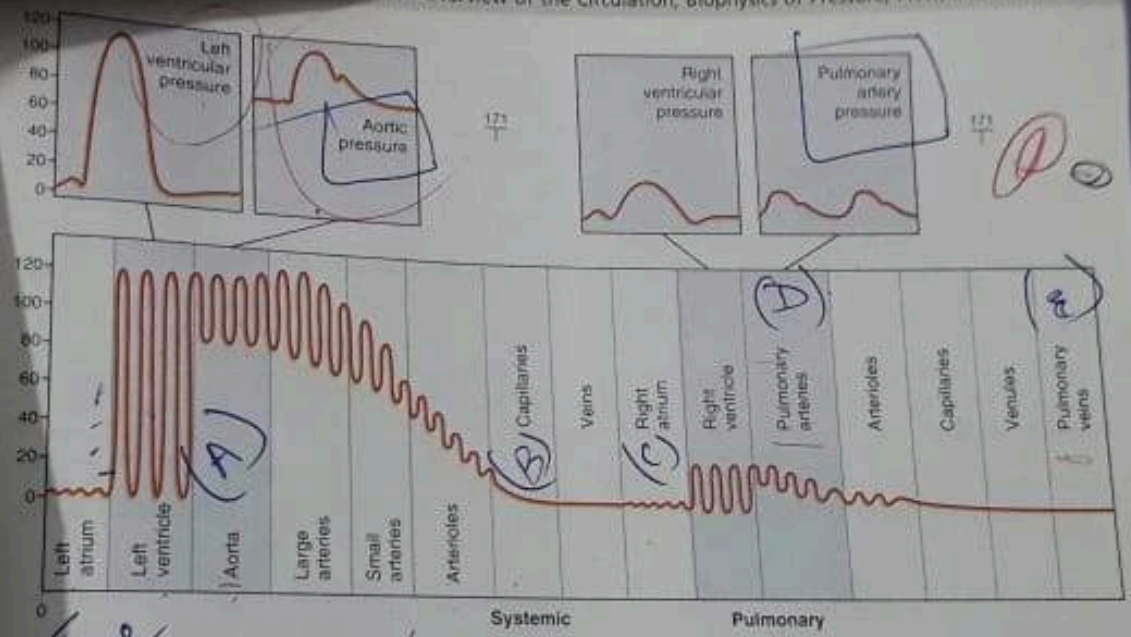


Figure 14-2 Normal blood pressures in the different portions of the circulatory system when a person is lying in the horizontal position.

tissue continuously monitor tissue needs, such as the availability of oxygen and other nutrients and the accumulation of carbon dioxide and other tissue waste products, and these microvessels in turn act directly on the local blood vessels, dilating or constricting them, to control local blood flow precisely to that level required for the tissue activity. Also, nervous control of the circulation from the central nervous system and hormones provide additional help in controlling tissue blood flow.

2. *Cardiac output is the sum of all the local tissue flows.* When blood flows through a tissue, it immediately returns to the heart and is pumped out again. Therefore, the amount of blood flowing through all tissues per minute—that is, the cardiac output—is determined almost entirely by the sum of all the local tissue blood flows.

Thus, the needs of the individual tissues are served specifically by the circulation. In the remainder of this chapter, we begin to discuss the basic details of the management of tissue blood flow and control of cardiac output and arterial pressure.

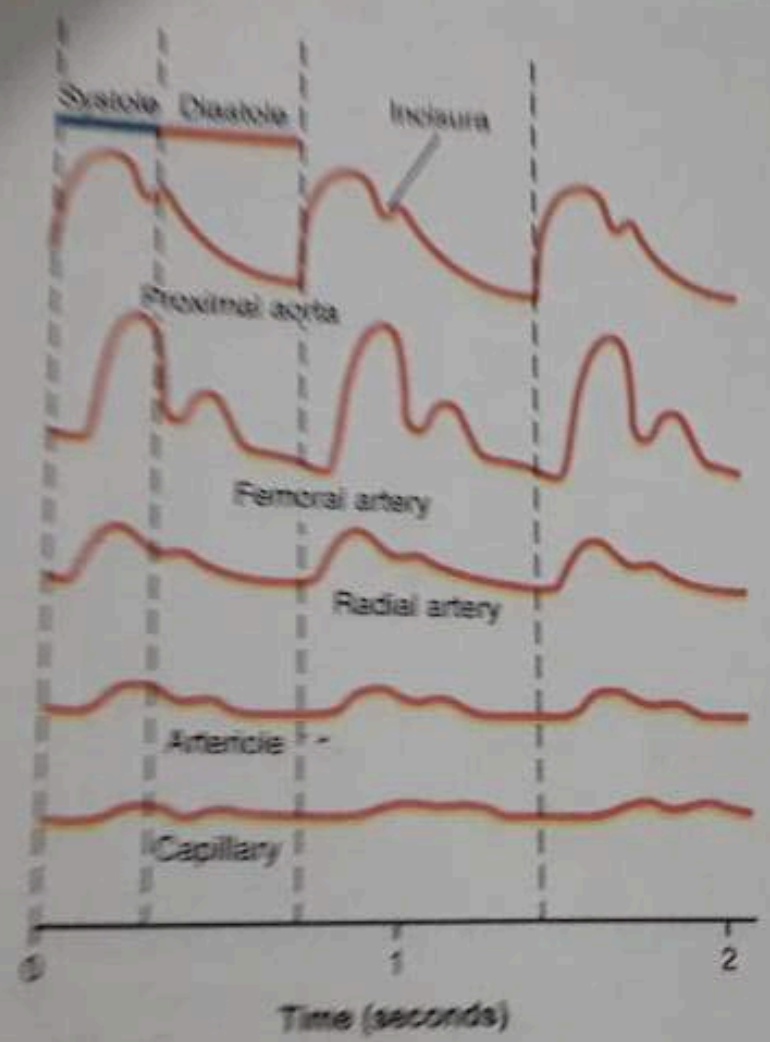


Figure 15-6. Changes in the pulse pressure contour as the pulse wave travels toward the smaller vessels.

special studies are necessary. Instead, the ... mines systolic and diastolic pressures ... means, usually by the *auscultatory method*.

Auscultatory Method. Figure 15-7 ... tory method for determining systolic and ... pressures. A stethoscope is placed over the ... artery and a blood pressure cuff is inflated ... upper arm. As long as the cuff continues to ... the arm with too little pressure to close ... artery, no sounds are heard from the ... with the stethoscope. However, when the cuff ... great enough to close the artery during part ... pressure cycle, a sound is then heard with ... These sounds are called *Korotkoff sounds*, ... Nikolai Korotkoff, a Russian physician who ... them in 1905.

The Korotkoff sounds are believed to be ... by blood jetting through the partly occluded ... by vibrations of the vessel wall. The jet causes ... in the vessel beyond the cuff, and this turbulence ... the vibrations heard through the stethoscope.

In determining blood pressure by the ... method, the pressure in the cuff is first ... above arterial systolic pressure. As long as the ... sure is higher than systolic pressure, the brachial ... remains collapsed so that no blood jets into ... artery during any part of the pressure cycle. ... Korotkoff sounds are heard in the lower arm ...



...gure 15-7). Blood ... meath the cuff du... ...e begins to hear ...tery in synchron... ...unds begin to be ...e manometer c... ...e systolic press... ...As the pressu... ...orotkoff sound... ...pping quality ...ality. Then, fa... ...ar diastolic p... ...uffled quality ...anometer pr... ...the muffled

main reason peripheral factors are usually so
ant in controlling cardiac output is that the heart
ilt-in mechanism that normally allows it to pump
ically whatever amount of blood that flows into
atrium from the veins. This mechanism, called
k-Starling law of the heart, was discussed in
Basically, this law states that when increased
of blood flow into the heart, the increased
tches the walls of the heart chambers. As
the stretch, the cardiac muscle contracts
ed force, and this action empties the extra
has entered from the systemic circulation.
e blood that flows into the heart is automati-
without delay into the aorta and flows again
irculation.

portant factor, discussed in Chapter 10, is
the heart causes the heart to pump faster,
increased heart rate. That is, stretch of the
e wall of the right atrium has a direct effect
city of the node to increase the heart rate
o 15 percent. In addition, the stretched
ates a nervous reflex called the *Bainbridge*
ex to the vasomotor center of the brain
o the heart by way of the sympathetic
lso to increase the heart rate.

rmal unstressed conditions, the cardiac
ed mainly by peripheral factors that

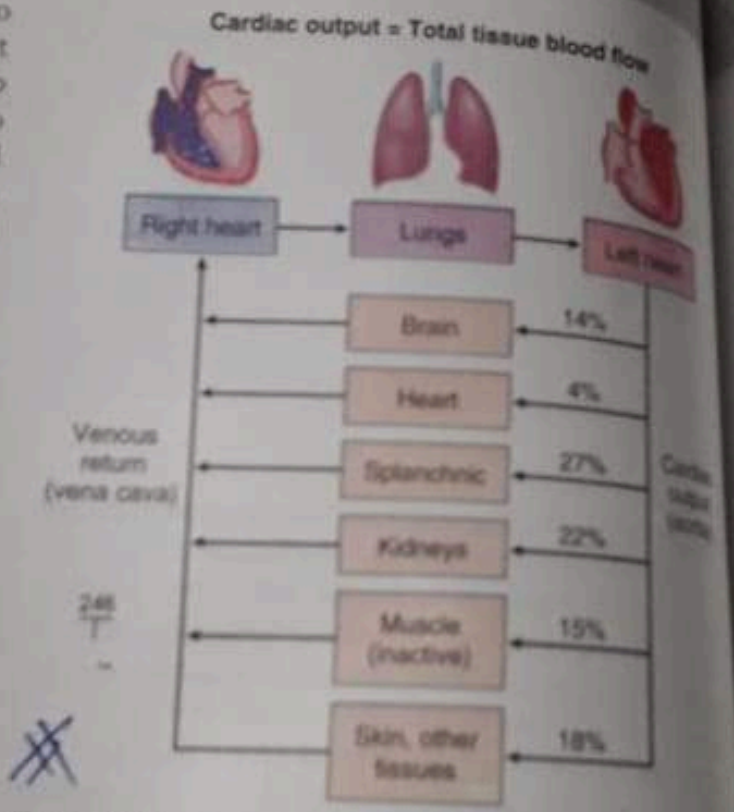


Figure 20-2. Cardiac output is equal to venous return and is the sum of tissue and organ blood flows. Except when the heart is severely weakened and unable to adequately pump the venous return, cardiac output (total tissue blood flow) is determined mainly by the metabolic needs of the tissues and organs of the body.

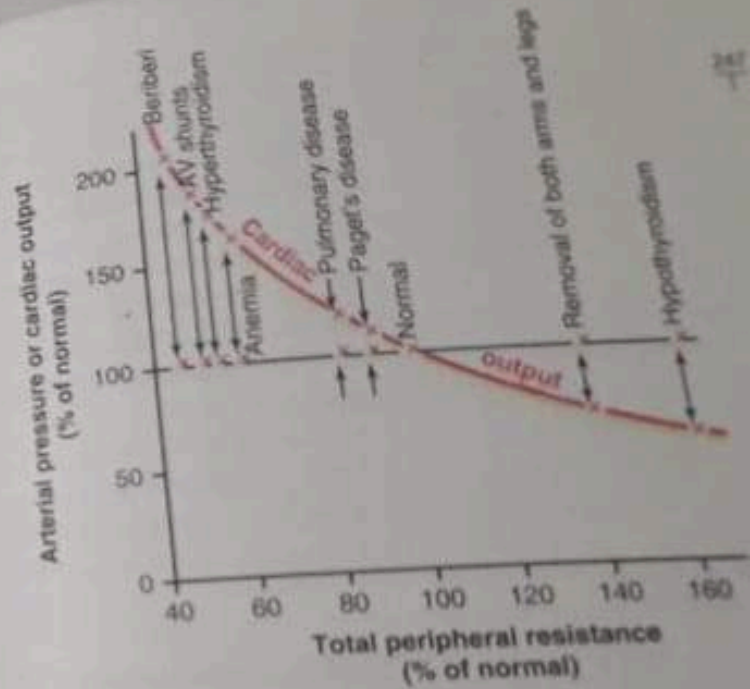


Figure 20-4. Chronic effect of different levels of total peripheral resistance on cardiac output, showing a reciprocal relationship between total peripheral resistance and cardiac output. Adapted from Guyton AC: Arterial Pressure and Hypertension. Philadelphia: WB Saunders, 1980.)

peripheral resistance increases above normal, the
output falls; conversely, when the total peripher
tance decreases, the cardiac output increases.

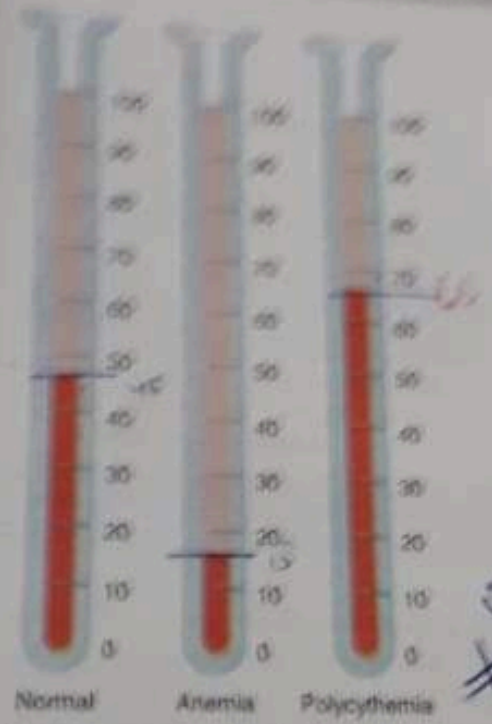


Figure 14-10. Hematocrits in a healthy (normal) person and in patients with anemia and polycythemia. The numbers refer to percentage of the blood composed of red blood cells.

removal of a kidney also removes a parallel circuit and reduces the total vascular conductance and total blood flow (i.e., cardiac output) while increasing total peripheral vascular resistance.

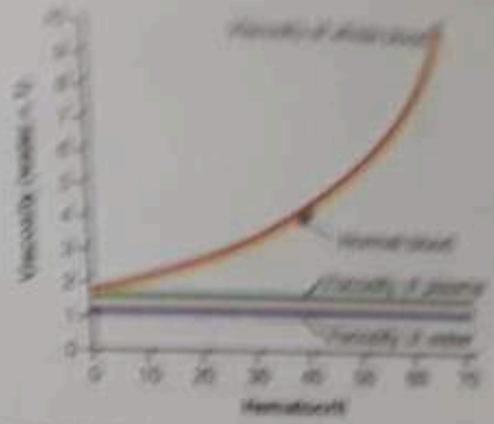


Figure 14-11. (Effect of hematocrit on blood viscosity (water viscosity = 1).

Increasing Hematocrit Markedly Increases Blood Viscosity. The viscosity of blood increases drastically as the hematocrit increases, as shown in Figure 14-11. The viscosity of whole blood at normal hematocrit is about 3 to 4, which means that three to four times as much pressure is required to force whole blood as to force water through the same blood vessel. When the hematocrit rises to 60 or 70, which it often does in persons with *polycythemia*, the blood viscosity can become as great as 10 times that of water, and its flow through blood vessels is greatly retarded.

Other factors that affect blood viscosity are the plasma

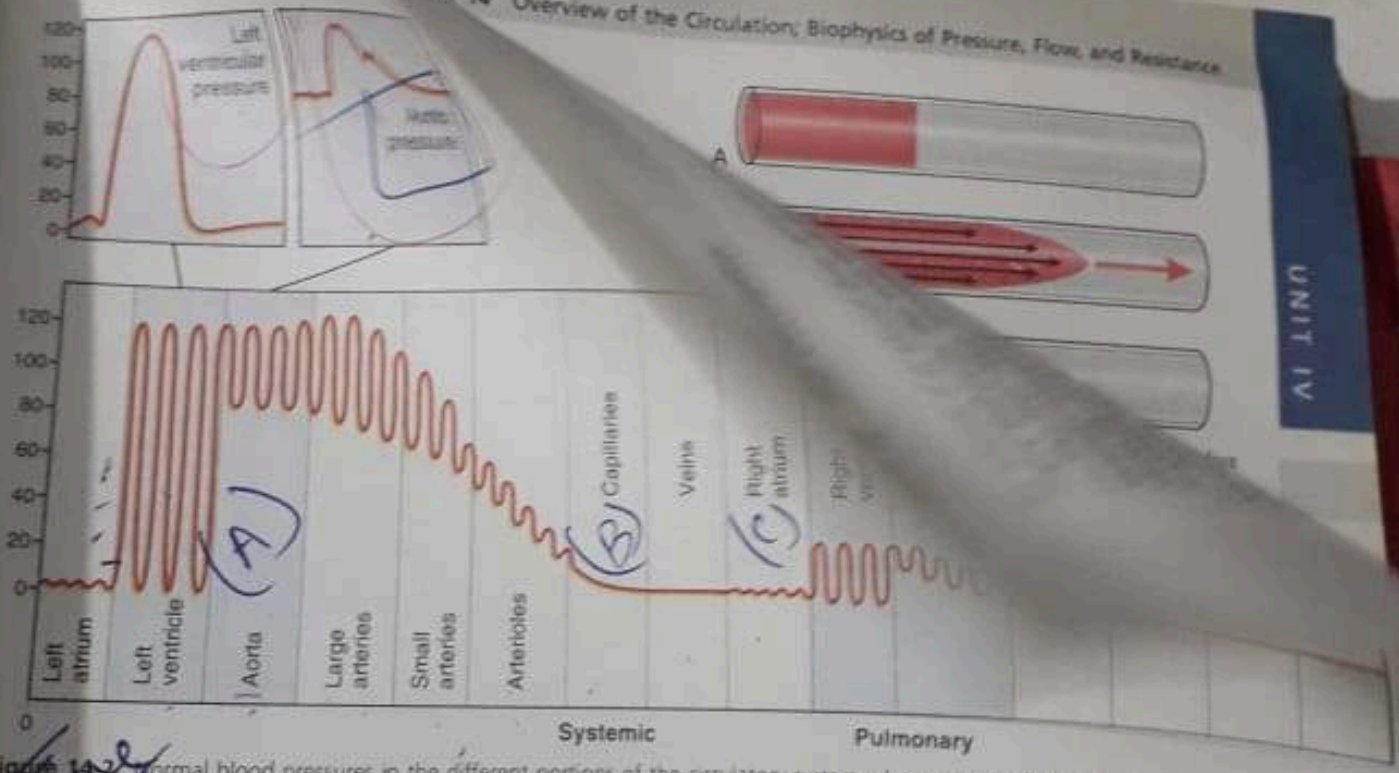


Figure 14-2 Normal blood pressures in the different portions of the circulatory system when a person is lying in the horizontal position.

tissue continuously monitor tissue needs, such as the availability of oxygen and other nutrients and the accumulation of carbon dioxide and other tissue waste products, and these microvessels in turn act directly on the local blood vessels, dilating or constricting them, to control local blood flow precisely to that level required for the tissue activity. Also

arterioles in many tissues so that more blood accumulates in the large arteries to increase the arterial pressure. Then, over more prolonged periods—hours and days—the kidneys play an additional major role in pressure control both by secreting pressure-controlling hormones and by regulating the blood volume.

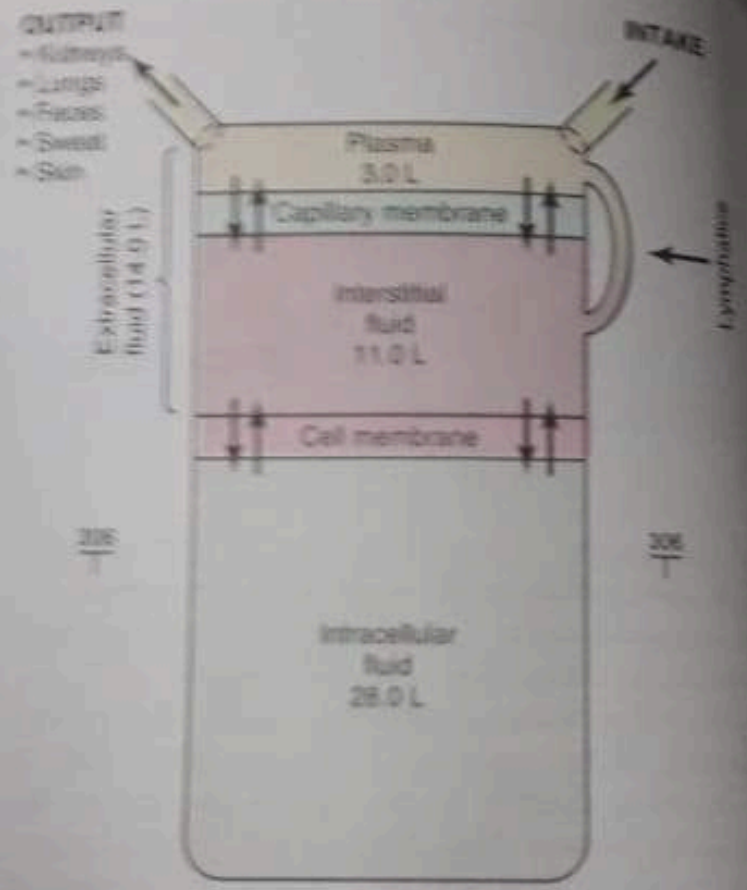


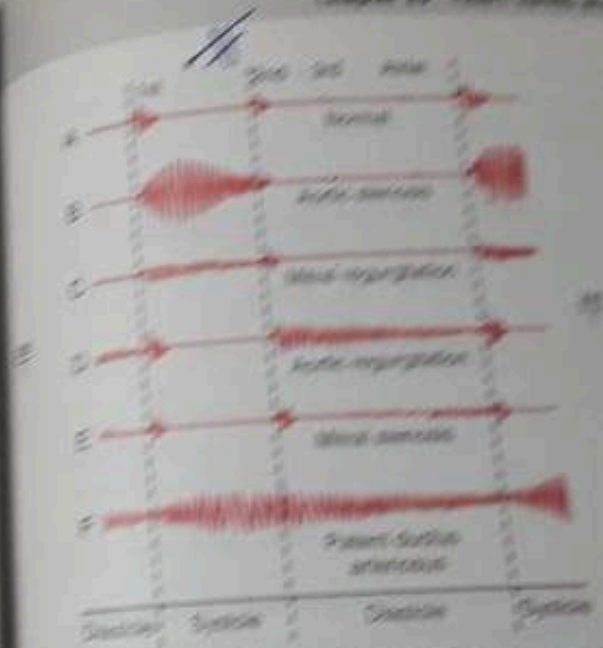
Figure 25-1. Summary of body fluid regulation, including the major body fluid compartments and the membranes that separate these compartments. The values shown are for an average 70-kilogram adult man.

The fluid of each cell contains its individual mixture of different constituents, but the concentrations of these substances are similar from one cell to another. In fact, the composition of cell fluids is remarkably similar even in different animals, ranging from the most primitive microorganisms to humans. For this reason, the intracellular fluid of all the different cells together is considered to be one large fluid compartment.

EXTRACELLULAR FLUID COMPARTMENT

All the fluids outside the cells are collectively called the *extracellular fluid*. Together these fluids account for about 20 percent of the body weight, or about 14 liters in a 70-kilogram man. The two largest compartments of the extracellular fluid are the *interstitial fluid*, which makes up more than three fourths (11 liters) of the extracellular fluid, and the *plasma*, which makes up almost one fourth of the extracellular fluid, or about 3 liters. The plasma is the noncellular part of the blood; it exchanges substances continuously with the interstitial fluid through the pores of the capillary membranes. These pores are highly permeable to almost all solutes in the extracellular fluid except the proteins. Therefore, the extracellular fluids are constantly mixing, so the plasma and interstitial fluids have about the same composition except for proteins, which have a higher concentration in the plasma.





and increased resistance to ventricular filling, for example, a fourth heart sound is often heard in older patients with left ventricular hypertrophy.

Chest Surface Areas for Auscultation of Heart Sounds. Listening to the sounds of the heart

third heart sound can be recorded in only one third to one half of all people, and the atrial heart sound can be recorded in perhaps one fourth of all people.

VALVULAR LESIONS

Rheumatic Valvular Lesions

METHODS FOR MEASURING AND DIASTOLIC PRESSURES

tical to use pressure recorders that requ
ion into an artery for making routine arteri
asurements in human patients, althoug
of recorders are used on occasion whe
s are necessary. Instead, the clinician dete
c and diastolic pressures through indire
y by the *auscultatory method*.

Method. Figure 15-7 shows the auscult
or determining systolic and diastolic arteri
stethoscope is placed over the antecubit
blood pressure cuff is inflated around th
s long as the cuff continues to compres
too little pressure to close the brachi
nds are heard from the antecubital arte
oscope. However, when the cuff pressur
o close the artery during part of the arter
a sound is then heard with each pulsati
are called *Korotkoff sounds*, named af
koff, a Russian physician who describ

off sounds are believed to be caused ma
g through the partly occluded vessel a
it causes turbule

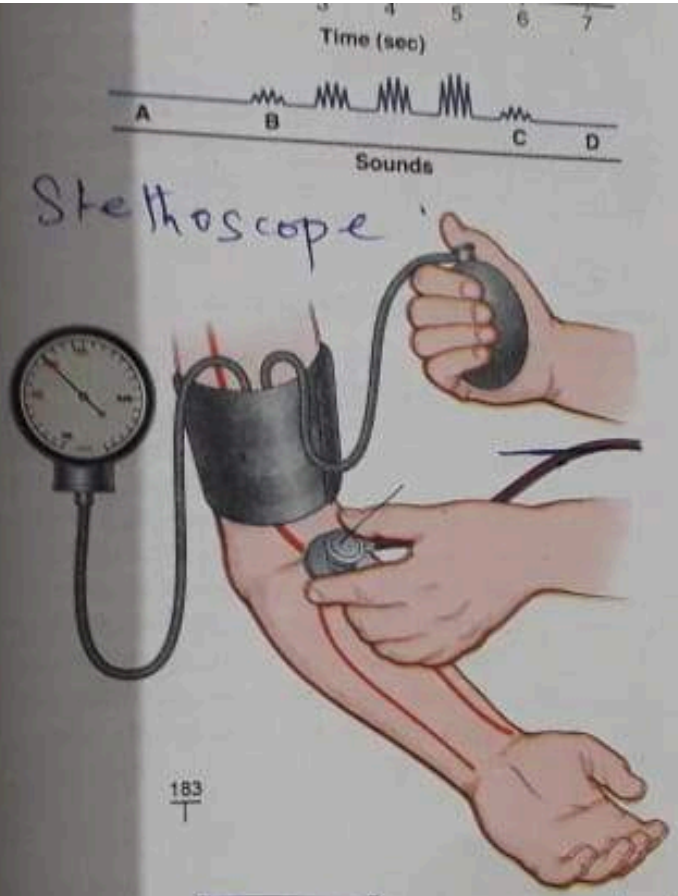


Figure 15-7. The auscultatory method for measuring systolic and diastolic arterial pressures.

Figure 15-7), blood begins to flow through the artery beneath the cuff during the peak of systolic pressure, and one begins to hear *tapping* sounds from the antecubital artery in synchrony with the heartbeat. As soon as these sounds begin to be heard, the pressure level indicated by the manometer connected to the cuff is about equal to

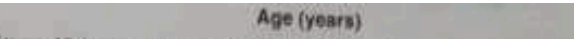


Figure 15-8. Changes in systolic, diastolic, and mean arterial pressures with age. The shaded areas show the approximate normal ranges.

the diastolic pressure determined by direct intra-arterial catheter. As the cuff pressure falls a few mm Hg further, the artery no longer closes during diastole, which means that the basic factor causing the sounds (the jetting of blood through a squeezed artery) is no longer present. Therefore, the sounds disappear entirely. Many clinicians believe that the pressure at which the Korotkoff sounds completely disappear should be used as the diastolic pressure, except in situations in which the disappearance of sounds cannot reliably be determined because sounds are audible even after complete deflation of the cuff. For example, in patients with arteriovenous fistulas for hemodialysis or with aortic insufficiency, Korotkoff sounds may be heard after complete deflation of the cuff.

The auscultatory method for determining systolic and diastolic pressures is not entirely accurate, but it usually gives values within 10 percent of those determined by direct catheter measurement from inside the arteries.

Normal Arterial Pressures as Measured by the Auscultatory Method. Figure 15-8 shows the approximate normal systolic and diastolic arterial pressures at different ages. The progressive increase in pressure with age results from the effects of aging on the blood pressure control mechanisms. We shall see in Chapter 19 that the kidneys are primarily responsible for this long-term regulation of arterial pressure; it is well known that the kidneys exhibit definitive changes with age, especially after the age of 50 years.

A slight extra increase in systolic pressure usually occurs beyond the age of 60 years. This increase results from decreasing distensibility, or "hardening," of the arter- which is often a result of *atherosclerosis*. The final

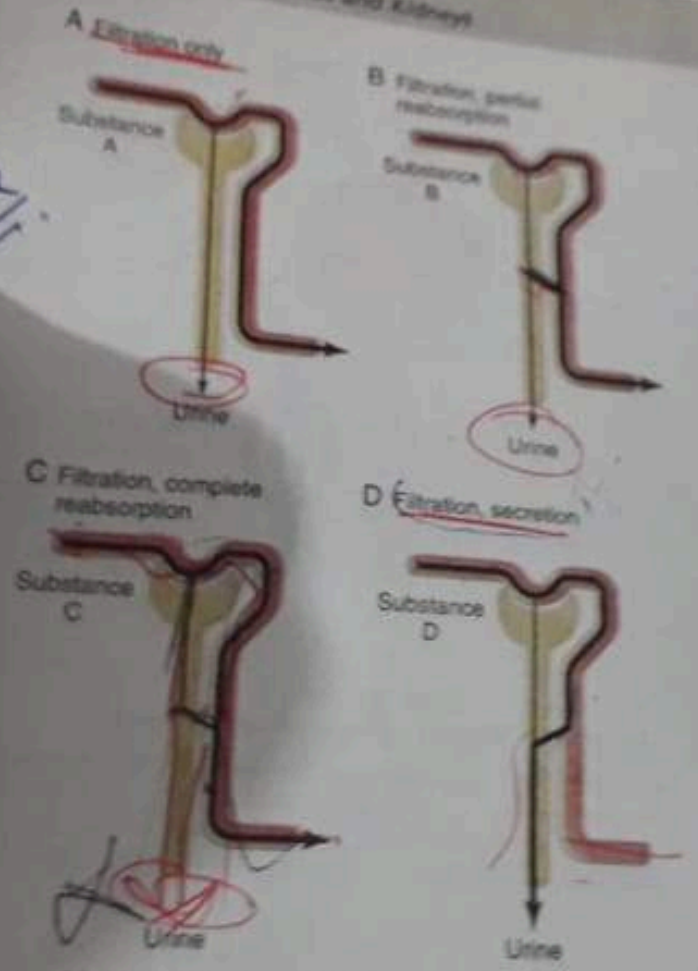


Figure 26-10. Renal handling of four hypothetical substances. *A*, The substance is freely filtered but not reabsorbed. *B*, The substance is freely filtered, but part of the filtered load is reabsorbed back in the blood. *C*, The substance is freely filtered but is not excreted in the urine because all the filtered substance is reabsorbed

as amino acids and glucose, are completely reabsorbed from the tubules and do not appear in the urine even though large amounts are filtered by the glomerular capillaries.

Each of the processes—glomerular filtration, tubular reabsorption, and tubular secretion—is regulated according to the needs of the body. For example, when the body has excess sodium in the body, the rate at which sodium is filtered usually increases and a smaller fraction of the filtered sodium is reabsorbed, causing increased excretion of sodium.

For most substances, the rates of filtration and reabsorption are extremely large relative to the rates of excretion. Therefore, even slight changes in filtration or reabsorption can lead to relatively large changes in renal excretion. For example, an increase in the glomerular filtration rate (GFR) of only 10 percent (from 180 to 198 L/day) would raise urine volume (if tubular reabsorption remained constant) from 1 L/day to 19.5 L/day! In reality, changes in glomerular filtration rate and tubular reabsorption usually act in a coordinated manner to produce the necessary changes in renal excretion.

Why Are Large Amounts of Solutes Filtered and Then Reabsorbed by the Kidneys?

One might question the wisdom of filtering such large amounts of water and solutes and then reabsorbing most of these substances. One advantage of this system is that it allows the kidneys to rapidly remove certain substances from the body that depend mainly on glomerular filtration for their excretion. Most waste products

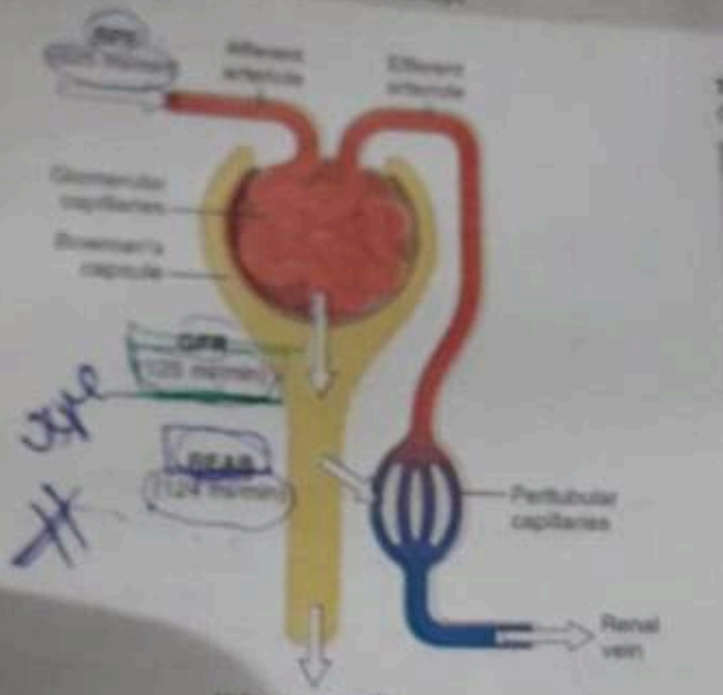


Figure 27-1. Average values for total renal plasma flow (RPF), glomerular filtration rate (GFR), and urine flow rate. RBF is equal to renal blood flow × (1 − Hematocrit). Note that GFR averages about 20% of the RPF, while urine flow rate is only about 1% of the GFR. Therefore, more than 99% of the fluid filtered is normally reabsorbed. The filtration fraction is the GFR/RPF.

GFR / day ⇒ 180 ml.

100% glucose A.P ⇒ 65Y

Proximal tubule

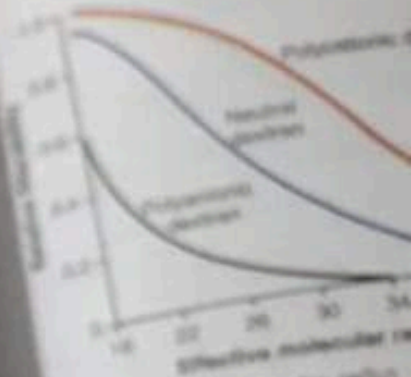
Table 27-1 Filterability of Substances in Glomerular Capillaries Based on Molecular Weight

Substance	Molecular Weight
Water	18
Sodium	23
Glucose	180
Inulin	5000
Myoglobin	17,000
Albumin	69,000

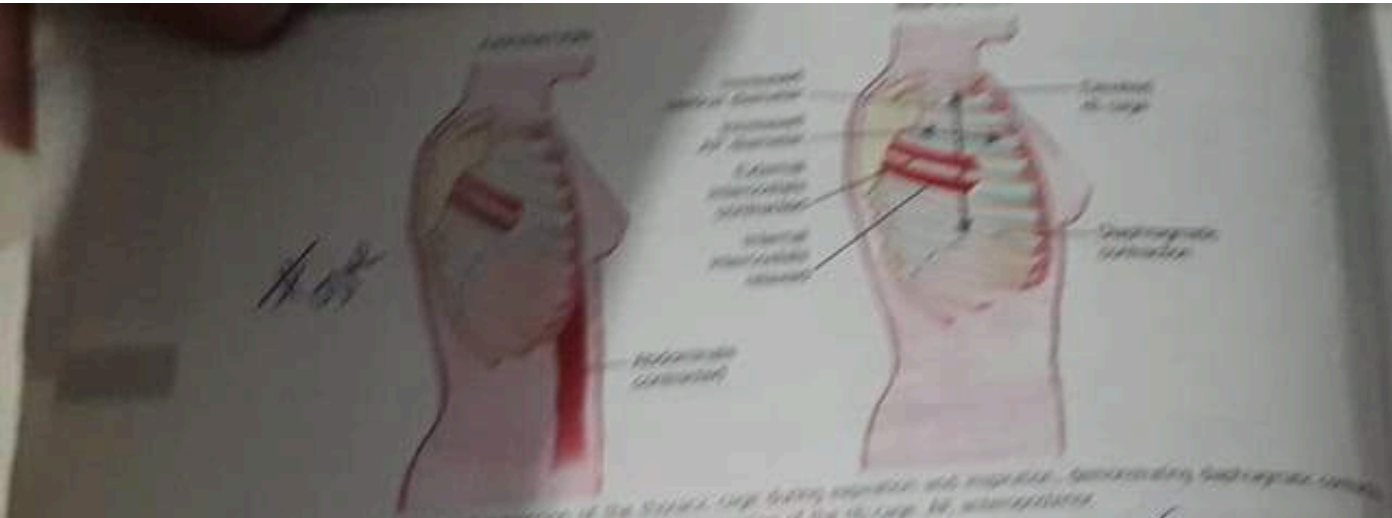
which the glomerular filtrate moves. Many of these proteins, which also have negative charges, pose a double restriction to filtration of plasma proteins. The negative charges of the glomerular capillary wall prevent the filtration of plasma proteins.

Filterability of Solutes Is Inversely Related to Their Size. The glomerular capillary membrane is much more permeable than most other capillaries, but it is also much less permeable and therefore filters fluid at a high rate. Despite the high glomerular filtration rate, the glomerular filtration barrier is highly selective in determining which molecules will be filtered, depending on their size and electrical charge.

Table 27-1 lists the effect of molecular size on the filterability of different molecules. A filterability of 1.0 means that the substance is filtered as freely as water. A filterability of 0.75 means that the substance is filtered only 75 percent as rapidly as water. Note that small molecules such as sodium and small organic compounds, including glucose, are freely filtered. As the molecular weight

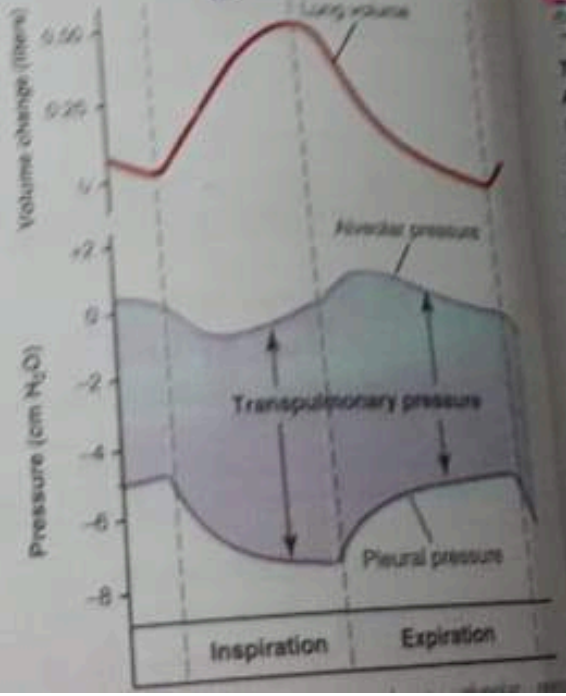


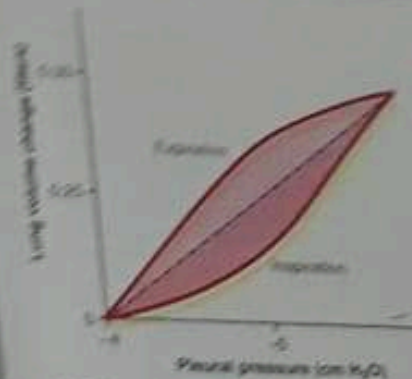
Effective molecular radius

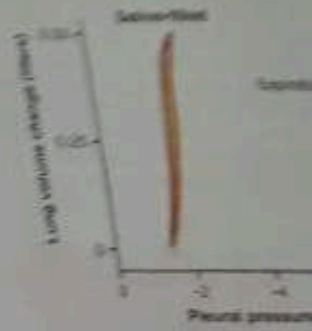


lungs within the cavity. Further, continual suction of excess fluid into lymphatic channels maintains a slight suction between the visceral surface of the lung pleura and the parietal pleural surface of the thoracic cavity. Therefore, the lungs are held to the thoracic wall as if glued there, except that they are well lubricated and can slide freely as the chest expands and contracts.

Pleural Pressure and Its Changes during Respiration. Pleural pressure is the pressure of the fluid in the thin capsular space between the lung pleura and the chest wall pleura. As noted earlier, this pressure is normally a slight suction, which means a slightly negative pressure. The normal pleural pressure at the beginning of inspiration is about −5 centimeters of water, which is the amount of suction required to hold the lungs open to their resting level. During normal inspiration, expansion of the chest cage pulls outward on the lungs with greater force and creates more negative pressure, to an average of about −7.5 cen-







...expired air out of the lungs during the 2 to 3 seconds of expiration.

Transpulmonary Pressure—The Difference between Alveolar and Pleural Pressures. Note in **Figure 37-2** that the *transpulmonary pressure* is the pressure difference between that in the alveoli and that on the outer surfaces of the lungs (pleural pressure), and it is a measure of the elastic forces in the lungs that tend to collapse the lungs at each instant of respiration, called the *recoil pressure*.

Compliance of the Lungs

The extent to which the lungs will expand for each unit increase in transpulmonary pressure (if enough time is allowed to reach equilibrium) is called the *lung compliance*. The total compliance of both lungs together in the normal adult human averages about 200 milliliters of air per centimeter of water transpulmonary pressure. That is, every time the transpulmonary pressure increases 1 centimeter of water, the lung volume, after 10 to 20 seconds, will expand 200 milliliters.

Compliance Diagram of the Lungs. Figure 38-3 is a

...tissue and (2) elastic forces caus... fluid that lines the inside walls ... air spaces.

The elastic forces of the ... mainly by elastin and colla... the lung parenchyma. In de... an elastically contracted a... lungs expand, the fibers b... thereby elongating and ex...

The elastic forces cau... more complex. The si... shown in Figure 38-4, ... diagram of the lungs w... when filled with air. ... there is an interface b... in the alveoli. In lung... no air-fluid interfac... effect is not present... tive in the lung fill...

Note that trans... air-filled lungs w... required to expa... one can conclu... to cause collap... about one thir...

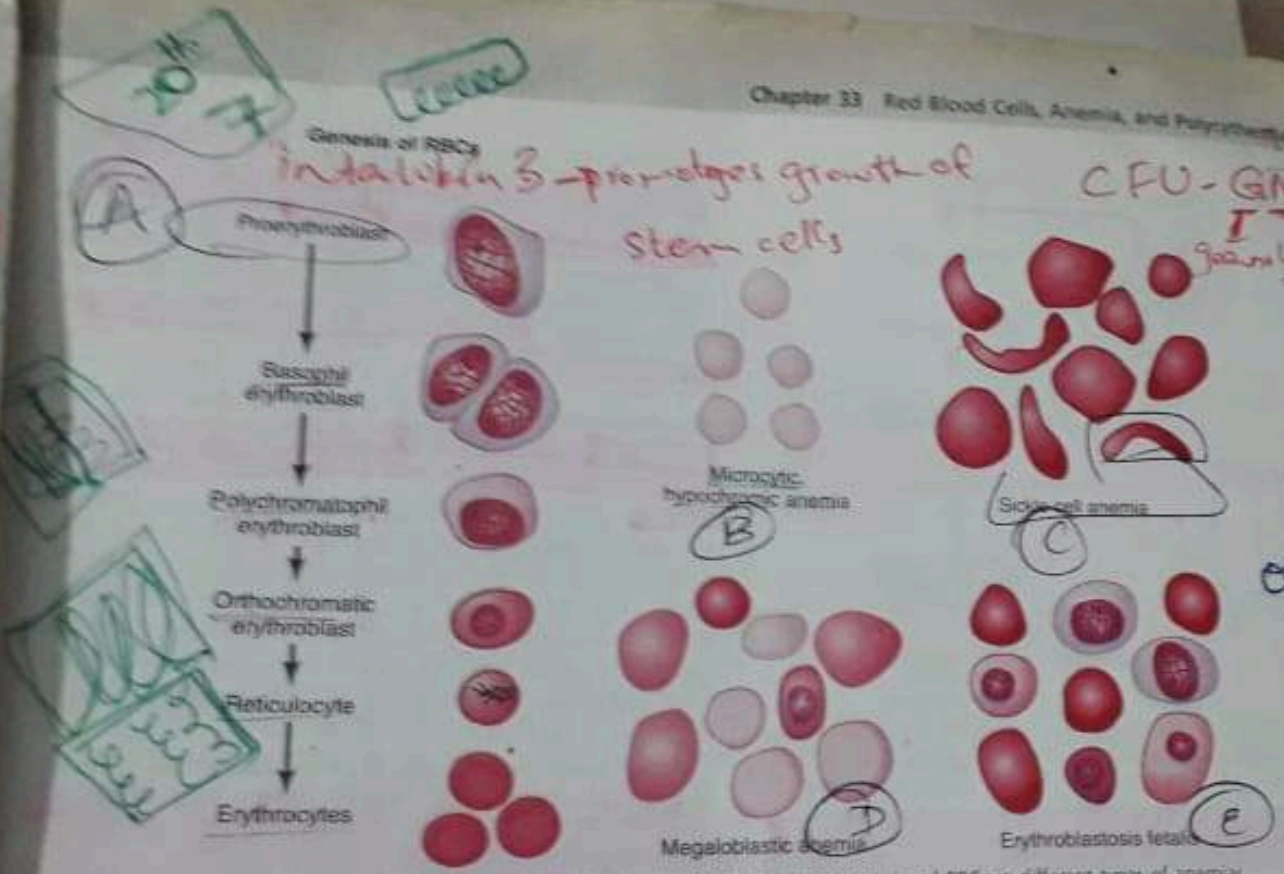


Figure 33-3. Genesis of normal red blood cells (*RBCs*) and characteristics of RBCs in different types of anemias.

granulocytes and monocytes have the designation *CFU-GM* and so forth.

Growth and reproduction of the different stem cells are controlled by multiple proteins called *growth inducers*. At least four major growth inducers have been described, each having different characteristics. One of these,

point in Figure 33-3. Under appropriate stimulation, large numbers of these cells are formed from the CFU-E stem cells.

Once the proerythroblast has been formed, it divides multiple times, eventually forming many mature RBCs. The first-generation cells are called *basophil erythroblasts* because they stain with basic dyes; the cell at this time

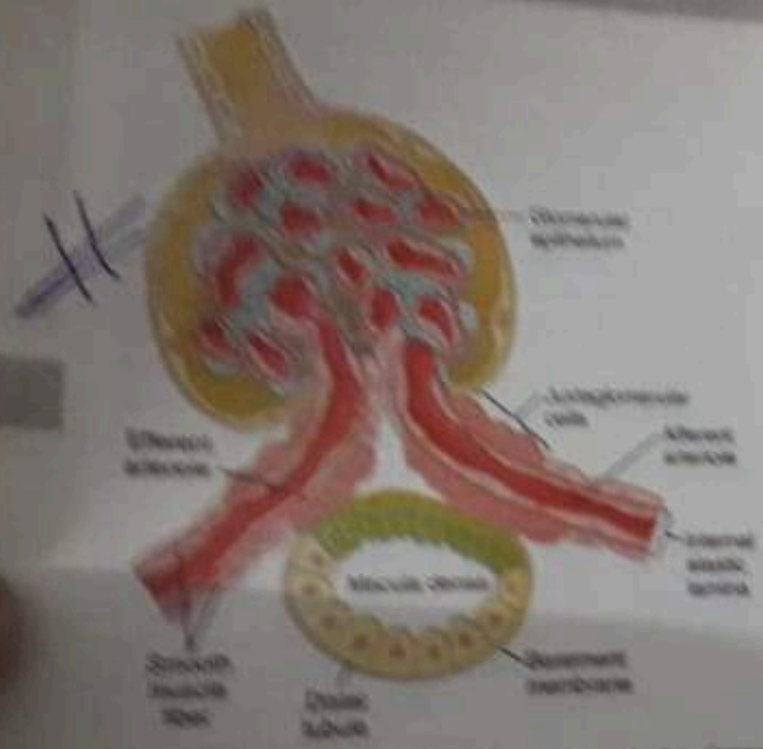


Figure 27-10. Structure of the juxtaglomerular apparatus, demonstrating its possible feedback role in the control of nephron function.

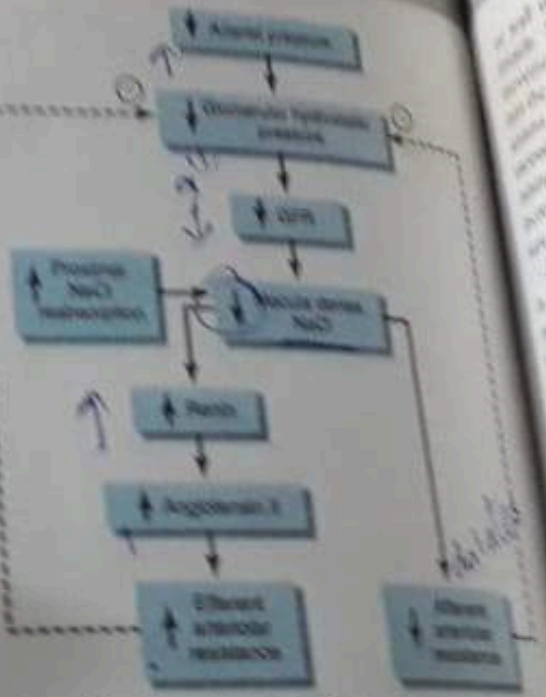


Figure 27-11. Macula densa feedback mechanism to autoregulation of glomerular hydrostatic pressure and glomerular filtration rate (GFR) during decreased renal arterial pressure.

Henle, causing increased reabsorption of the percentage of sodium and chloride ions delivered to the ascending loop of Henle, thereby reducing the concentration of sodium chloride at the macula densa cells. This decrease

...ed by contraction of the vascular smooth ... stretch of the vascular wall allows increased ... of calcium ions from the extracellular fluid ... the cells, causing them to contract through the mech... discussed in Chapter 8. This contraction preven... stretch of the vessel and at the same time, ... vascular resistance, helps prevent excess... in renal blood flow and GFR when arterial p...

...though the myogenic mechanism probably ope... ...arterioles throughout the body, its importa... ...blood flow and GFR autoregulation has been ... by some physiologists because this pr... ...mechanism has no means of directly de... ...in renal blood flow or GFR per se. On th... ...this mechanism may be more important in ... ...kidney from hypertension-induced ... ...to sudden increases in blood pressure ... ...vascular response in afferent arteriol... ...vessels and therefore attenuates trans... ...arterial pressure to the glomerular ...

Other Factors That Increase Renal Blood ... ...gh Protein Intake and Increased Bl...

...though renal blood flow and GFR are reg... ...ous conditions, there are circumstan... ...ous variables change significantly. For ex... ...protein intake is known to increase both r... ...and GFR. With a long-term high-protein d... ...to consume large amounts of meat, the r... ...al renal blood flow are due partly t... ...times however, GFR and renal blood ... ...to 30 percent within 1 or 2 hours af... ...high-protein meal.

...postrenal constrictor action of angiotensin II on efferent arterioles helps prevent serious reductions in glomerular hydrostatic pressure and GFR when renal perfusion pressure falls below normal. Administration of drugs that block

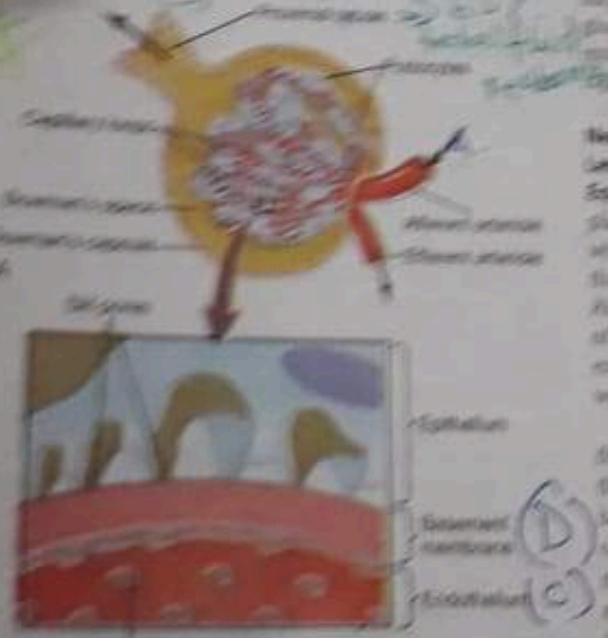


Figure 27-2. *A*, Basic structure of the glomerular capillaries. *B*, Cross section of the glomerular capillary membrane and its major components, including endothelium, basement membrane, and epithelium.

Negatively Charged Large Molecules Are Less Easily Than Positively Charged Molecules of Equal Molecular Size. The molecular [illegible] plasma protein albumin is only about [illegible] whereas the pores of the glomerular membrane are thought to be about 8 nanometers in radius. Albumin is restricted from filtration [illegible] of its negative charge and the electrostatic repulsion exerted by negative charges of the glomerular capillary wall proteoglycans.

Figure 27-3 shows how electrical charge affects the filtration of different molecular weight dextrans by the glomerulus. Dextrans are polysaccharides that can be manufactured as neutral molecules or with negative or positive charges. Note that for any given molecular radius, positively charged molecules are filtered much more readily than are negatively charged molecules. Neutral dextrans are also filtered more easily than are negatively charged dextrans of equal molecular radius. The reason for these differences in filtration is that the negative charges of the basement membrane and podocytes provide an important means for restricting

DETERMINANTS OF THE GFR

The GFR is determined by (1) the sum of the hydrostatic and colloid osmotic forces across the glomerular membrane, which gives the net filtration pressure, and (2) the glomerular K_f. Expressed mathematically, the GFR is the product of K_f and the net filtration pressure:

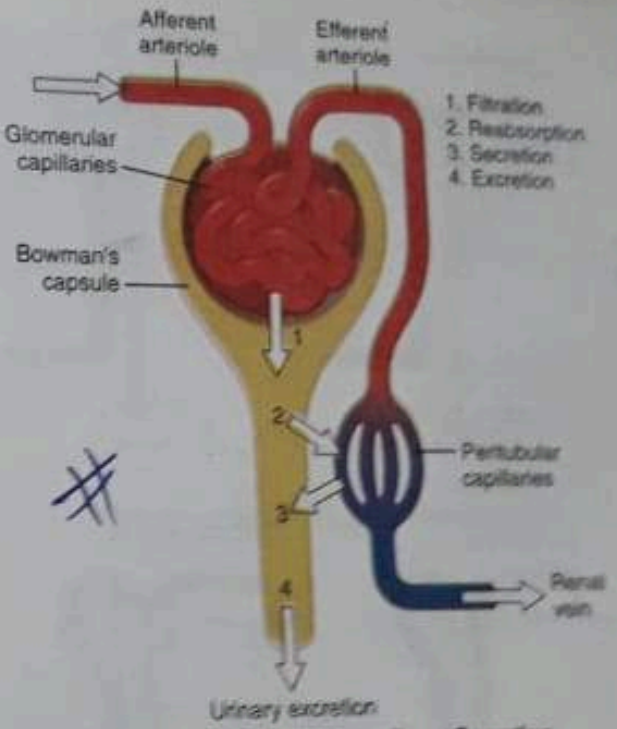
$$GFR = K_f \times \text{Net filtration pressure}$$

The net filtration pressure represents the sum of the hydrostatic and colloid osmotic forces that either promote or oppose filtration across the glomerular capillaries (Table 27-4). These forces include (1) hydrostatic pressure in the glomerular capillaries (glomerular hydrostatic pressure, P_G), which promotes filtration; (2) the hydrostatic pressure in Bowman's capsule (P_B) outside the glomerular capillaries, which opposes filtration; (3) the colloid osmotic pressure of the glomerular capillary plasma proteins (π_G), which opposes filtration; and (4) the colloid osmotic pressure of the

still intact, typical micturition reflexes can still occur. However, they are no longer controlled by the brain. During the first few days to several weeks after the damage to the cord has occurred, the micturition reflexes are suppressed because of the state of "spinal shock" caused by the sudden loss of facilitative impulses from the brain stem and cerebrum. However, if the bladder is emptied periodically by catheterization to prevent bladder injury caused by overstretching of the bladder, the excitability of the micturition reflex gradually increases until typical micturition reflexes return; then, periodic (but unannounced) bladder emptying occurs.

Some patients can still control urination in this condition by stimulating the skin (scratching or tickling) in the genital region, which sometimes elicits a micturition reflex.

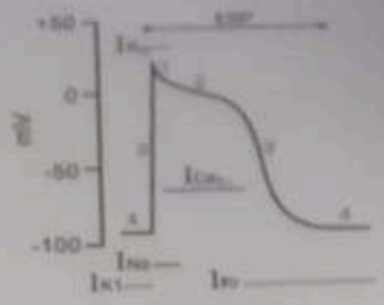
Uninhibited Neurogenic Bladder Caused by Lack of Inhibitory Signals from the Brain. Another abnormality of micturition is the so-called *uninhibited neurogenic bladder*, which results in frequent and relatively uncontrolled micturition. This condition derives from partial damage in the spinal cord or the brain stem that interrupts most of the inhibitory signals. Therefore, facilitative impulses passing continually down the cord keep the sacral centers so excitable that even a small quantity of urine elicits an uncontrollable micturition reflex, thereby pro-



Excretion = Filtration – Reabsorption + Secretion

Figure 26-9. Basic kidney processes that determine the composition of the urine. Urinary excretion rate of a substance is equal to the rate at which the substance is filtered minus its reabsorption rate plus the rate at which it is secreted from the peritubular capillary blood into the tubules.

products in the body, such as creatinine, are handled by



Q1. Phase 0 represents?

a) Influx of sodium
b) Closer of sodium channel
c) Influx of potassium channel
d) Closer of potassium channel
e) Opening of calcium channel

Q2. Phase 1 represents?

a) Influx of sodium
b) Partial repolarization
c) Influx of potassium channel
d) Closer of potassium channel
e) Opening of calcium channel

Q3.Phase 2 represents?

a) Influx of sodium
b) Partial repolarization
c) Influx of potassium channel
d) Closer of potassium channel
e) Opening of calcium channel

Q4.Phase 3 represents?

a) Influx of sodium
b) Partial repolarization
c) Influx of potassium channel
d) Closer of potassium channel
e) Opening of potassium channel

Q5.Phase 4 represent?

a) Influx of sodium
b) Partial repolarization
c) Influx of potassium channel
d) Closer of potassium channel
e) Resting membrane potential



Q26: Above mentioned condition occurs in

a) Cushing syndrome
b) Gigantism
c) Diabetes mellitus
d) Hypoparathyroidism
e) Hyperparathyroidism

Q27: This condition is due to

a) Acidosis
b) Depression of nervous system
c) Depression of sympathetic nervous system
d) Hypocalcemia
e) Hypercalcemia

Q28: Active form of Vitamin D is

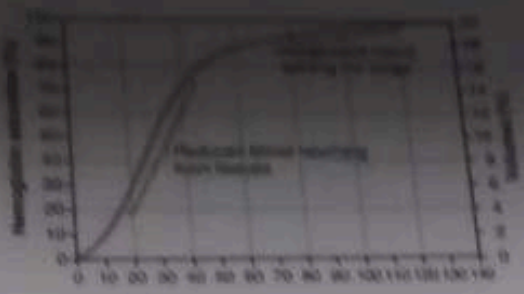
a) Cholcalciferol
b) 7 dehydrocholesterol
c) 25 hydroxycholcalciferol
d) 1,25dihydroxycholcalciferol
e) 1,25,26trihydroxycholcalciferol

Q29: 1,25dihydrocholcalciferol is formed in

a) Liver
b) Skin
c) Kidney
d) Intestine
e) Spleen

Q30: In which part of the nephron conversion of 25 hydroxycholcalciferol to 1,25dihydroxycholcalciferol take place

a) Glomerulus
b) Proximal convoluted tubules
c) Loop of Henle
d) Distal convoluted tubules
e) Collecting duct



Q.11: Right ward shift of the curve indicate

a) Haldane effect
b) Bohr's effect
c) Oxygen binds with haemoglobin
d) Low PCO2
e) Decreased 2,3 DPG

Q.12: Normal venous blood returning from the peripheral tissues, the P_{O_2} is about 40 mm Hg, and the saturation of haemoglobin averages

a) 95 %
b) 85 %
c) 75 %
d) 65 %
e) 55%

Q.13: If an adult man has 15gm haemoglobin / dl of blood the oxygen binding capacity in 100 ml of blood is

a) 1.34 ml of oxygen
b) 10.34 ml of oxygen
c) 15.01 ml of oxygen
d) 20.1 ml of oxygen
e) 21.1 ml of oxygen

Q.14: left ward shift of oxygen haemoglobin dissociation curve in

a) Increased body temperature
b) Decreased PO2
c) Increased 2,3 DPG
d) Acidosis
e) Alkalosis

Q.15: Normal partial pressure of oxygen in systemic arterial blood is

a) 40 mmHg
b) 50 mmHg
c) 87 mmHg
d) 95mmHg
e) 104 mmHg

Q2: What is second heart sound?

A: Vibration setup due to the closure of Semilunar Valves is called second heart sound.

Q3: What are the parts of conducting system of heart?

A: SA node, AV node, Bundle of His, internodal pathways, bundle branches, perkinje fibers.

Q4: What is "ST" segment and its duration in ECG?
A: 'S-T' segment is the time interval between the end of 'S' wave and the onset of 'T' wave. It is an isoelectric period. Normal duration of 'S-T' segment is 0.08 second.

Q5: What are unipolar limb leads?
A: These leads are also called augmented limb leads. The active electrode is connected to one of the limbs while indifferent electrode is obtained by connecting the other two limbs through a resistance. aVR, aVL, aVF.

A: The rate of formation of glomerular filtrate through glomerulus is called glomerular filtration rate

Q2: How much % of the tubular fluid is reabsorb through proximal convoluted tubules?

A: About 65% of the tubular fluid is reabsorbed through PCT.

Q3: Which structure is called counter current exchanger?

A: Vasa Recta

Q4: What are the structure which facilitate the water reabsorption through collecting ducts of nephron?

A: Aquaporin 2 under the influence of ADH

Q5: Which hormone secreted through juxtaglomerular apparatus?

A: Renin is the hormone secreted by JG cells of juxta glomerular apparatus.

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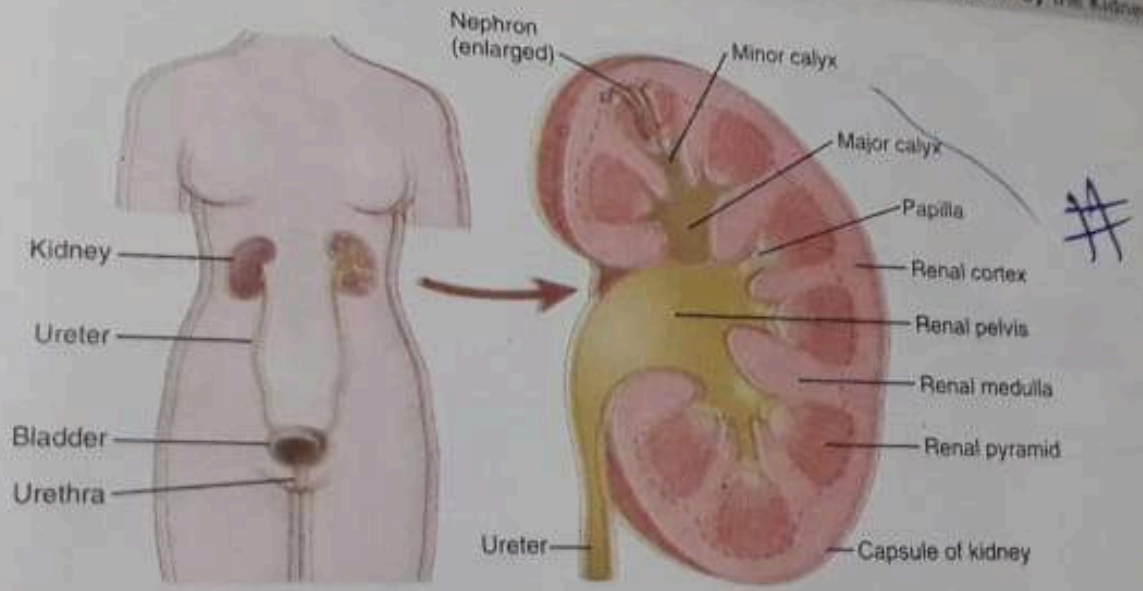


Figure 26-2. General organization of the kidneys and the urinary system.

RENAL BLOOD SUPPLY

Blood flow to ... kidneys is normally about 22 percent of ... put, or 1100 ml/min. The renal artery ... through the hilum and then ... form the *interlobar arteries*,

THE NEPHRON IS THE FUNCTIONAL UNIT OF THE KIDNEY

Each human kidney contains about 800,000 to 1,000,000 *nephrons*, each of which is capable of forming urine. The kidney cannot regenerate new nephrons. Therefore, with ... injury, disease, or normal aging, the number of

CHARACTERISTICS OF WHOLE MUSCLE CONTRACTION

Many features of muscle contraction can be demonstrated by eliciting single muscle twitches. This can be accomplished by instantaneous electrical excitation of the nerve to a muscle or by passing a short electrical stimulus through the muscle itself, giving rise to a single, sudden contraction lasting a fraction of a second.

Contractions Do Not Shorten Muscle, Isotonic Contractions Shorten Muscle at a

Muscle contraction is said to be iso- does not shorten during contrac- does shorten but the tension on ant throughout the contraction.

two types of muscle contraction

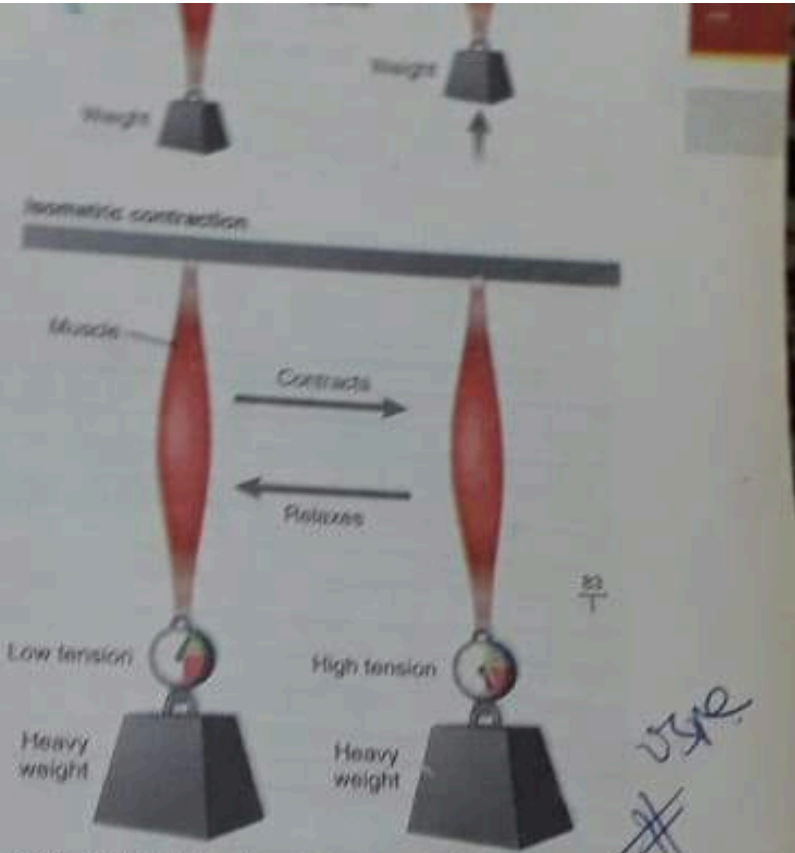


Figure 6-12. Isotonic and isometric systems for recording muscle contractions. Isotonic contraction occurs when the force of the muscle contraction is greater than the load and the tension on the muscle remains constant during the contraction; when the muscle contracts, it shortens and moves the load. Isometric contraction occurs when the load is greater than the force of the muscle contraction; the muscle creates tension when it contracts, but the overall length of the muscle does not change.

isometric system is often used when comparing the functional characteristics of different muscle types